



*Nazariy tur: 12-iyul 2011-yil
1-masala, 1-sahifa (jami 3 tadan)*

1. Uch jism masalasi va LISA

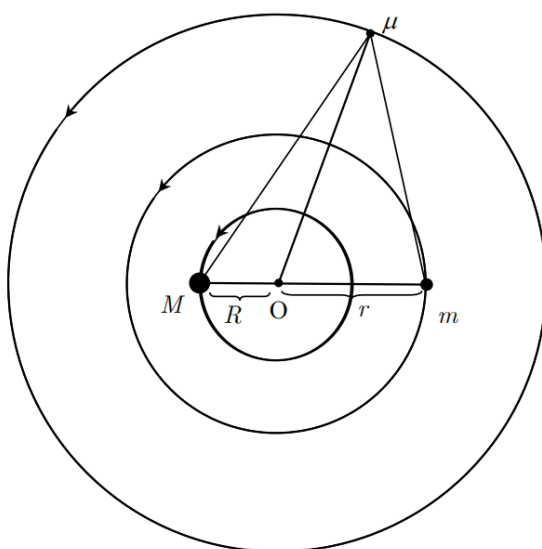


FIGURE 1 Coplanar orbits of three bodies.

.png

Figure 1: 1-rasm: Uch jismning bitta tekislikdagi orbitalari.

1.1 Ikkita gravitatsiyalashuvchi M va m massalar mos ravishda R va r radiusli aylanma orbitalar bo'ylab o'zlarining umumiy massalar markazi atrofida aylanmoqda. M va m jismlarni tutashtiruvchi to'g'ri chiziqlarning burchakli tezligi ω_0 ni R, r, M, m va univversal gravitatsiya doimiysi G orqali ifodalang. (1,5 ball)

1.2 Cheksiz kichik μ massali uchinchil jism xuddi o'sha massalar markazi atrofida, xuddi o'sha tekislikdagi aylanma orbita bo'ylab shunday harakatlanadiki, natijada μ jism ham M ga, ham m jismga nisbatan 1-rasmda ko'rsatilganidek qo'zg'almas (statsionar) bo'lib qoladi. Ushbu cheksiz kichik jism M va m jismlar bilan bir to'g'ri chiziqda yotmaydi deb hisoblang. Quyidagi parametrlarning qiymatlarini R va r orqali toping: (3,5 ball)

1.2.1 μ jismdan M jismgacha bo'lgan masofa.

1.2.2 μ jismdan m jismgacha bo'lgan masofa.

1.2.3 μ jismdan massalar markazigacha bo'lgan masofa.

1.3 $M = m$ bo'lgan holatni ko'rib chiqing. Agar μ jismga kichik radial g'alayonlanish ($O\mu$ chizig'i bo'ylab) berilsa, μ jismning g'alayonlanmagan vaziyati atrofida tebranishlarining burchakli chastotasi ω_0 orqali nimaga teng bo'ladi? μ jismning burchakli impulsi (fazoviy momenti) saqlanadi deb hisoblang. (3,2 ball)

Laser Interferometry Space Antenna (LISA) — bu past chastotali gravitatsion to'lqinlarni aniqlash (deteksiya qilish) uchun mo'ljallangan uchta bir xil kosmik apparatdan iborat guruhdir. Kosmik apparatlarning har biri 2-rasm va 3-rasmda ko'rsatilganidek, teng tomonli uchburchakning uchlarida joylashgan. Tomonlari (yoki "yelkalari") taxminan 5,0 million kilometr uzunlikka ega. LISA yulduz turkumi Quyosh atrofida, Erga o'xshash orbita bo'ylab, Erdan 20° burchakka ortda qolib harakatlanadi. Ularning har biri Quyosh atrofida biroz og'ishgan alohida orbita bo'ylab harakat qiladi. Amalda, ushbu uchta kosmik apparat o'zlarining umumiy markazi atrofida yiliga bir marta aylanma harakat qilayotgandek ko'rinadi.

Ular doimiy ravishda bir-birlari bilan lazer signallarini uzatadi va qabul qiladi. Umu-man olganda, uklar interferometrik usullar yordamida yelka uzunliklaridagi juda kichik o'zgarishlarni o'lchash orqali gravitatsion to'lqinlarni qayd etadi. Yaqin atrofdagi galaktikalardagi qora tuynuklar kabi ulkan massali jism (obyekt)larning to'qnashuvi gravitatsion to'lqinlarning manbalariga misol bo'la oladi.

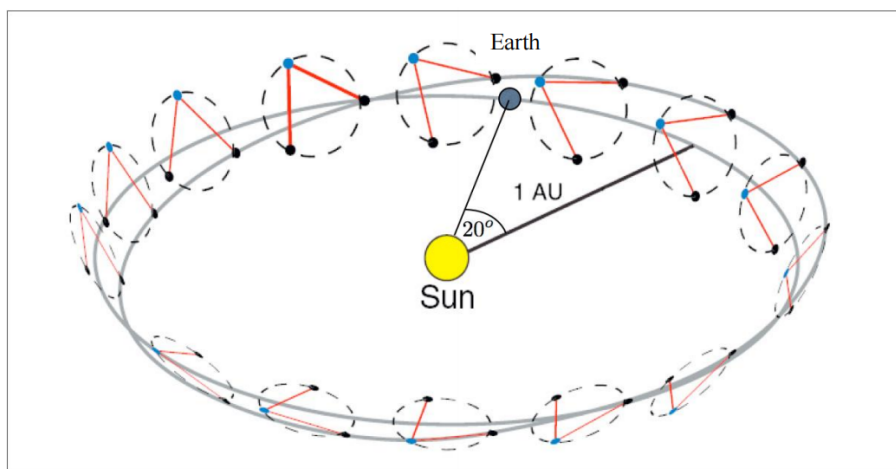


Figure 2: 2-rasm: LISA orbitasining tasviri. Uchta kosmik apparat o'zlarining massalar markazi atrofida 1 yil davrlik bilan aylanadi. Dastlab, ular Erdan 20° ortda qoladi. (Rasm muallifi D.A. Shaddock, "An Overview of the Laser Interferometer Space Antenna", *Publications of the Astronomical Society of Australia*, 2009, 26, pp.128-132.)

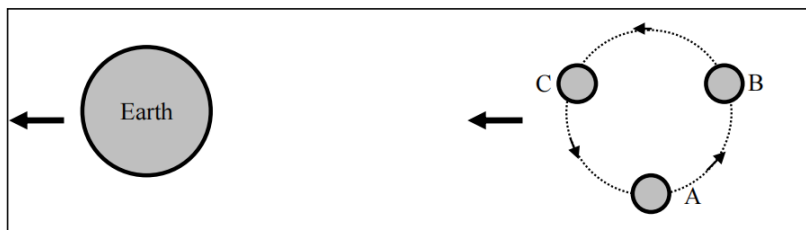


FIGURE 3 Enlarged view of the three spacecraft trailing the Earth. A, B and C are the three spacecraft at the corners of the equilateral triangle.

Figure 3: 3-rasm: Erdan keyin kelayotgan uchta kosmik apparatning kattalashtirilgan ko'rinishi. A, B va C — teng tomonli uchburchak uchlarida joylashgan uchta kosmik apparat.

1.4 Uchta kosmik apparat joylashgan tekislikda, bir kosmik apparatning boshqasiga nisbatan nisbiy tezligi qancha bo'ladi? (1,8 ball)

*Nazariy tur: 12-iyul 2011-yil
2-masala, 1-sahifa (jami 2 tadan)*

2. Zaryadlangan sovun pufakchasi

Ichki havo zichligi ρ_i , harorati T_i va radiusi R_0 bo'lgan sferik sovun pufakchasi zichligi ρ_a , atmosfera bosimi P_a va harorati T_a bo'lgan havo bilan o'ralgan. Sovun pardasi sirt tarangligi γ , zichligi ρ_s va qalinligi t ga ega. Sovunning massasi va sirt tarangligi harorat bilan o'zgarmaydi deb hisoblang. $R_0 \gg t$ deb hisoblang.

Sovun-havo chegarasining sirt yuzasini dA miqdorga oshirish uchun zarur bo'lgan dE energiya ortishi $dE = \gamma dA$ tenglama bilan beriladi, bu yerda γ — parda sirt tarangligi.

2.1 $\frac{\rho_i T_i}{\rho_a T_a}$ nisbatni γ , P_a va R_0 orqali ifodalang. (1,7 ball)

2.2 $\gamma = 0,0250 \text{ N} \cdot \text{m}^{-1}$, $R_0 = 1,00 \text{ cm}$ va $P_a = 1,013 \times 10^5 \text{ N} \cdot \text{m}^{-2}$ qiymatlardan foydalanib, $\frac{\rho_i T_i}{\rho_a T_a} - 1$ ning son qiymatini toping. (0,4 ball)

2.3 Pufakcha dastlab ichida iliqroq havo bilan hosil qilingan. Pufakcha harakatsiz havoda muallaq turishi (suzishi) uchun T_i ning minimal son qiymatini toping. $T_a = 300 \text{ K}$, $\rho_s = 1000 \text{ kg} \cdot \text{m}^{-3}$, $\rho_a = 1,30 \text{ kg} \cdot \text{m}^{-3}$, $t = 100 \text{ nm}$ va $g = 9,80 \text{ m} \cdot \text{s}^{-2}$ qiymatlardan foydalaning. (2,0 ball)

Pufakcha hosil bo'lgandan keyin bir muncha vaqt o'tgach, u atrof-muhit bilan issiqlik muvozanatida bo'ladi. Harakatsiz havodagi ushbu pufakcha tabiiy ravishda yerga qarab tushadi.

2.4 Issiqlik muvozanatida bo'lgan pufakchaning pastga tushishini to'xtatib turadigan yuqoriga yo'nalgan havo oqimining minimal u tezligini toping. Javobingizni ρ_s , R_0 , g , t va havoning qovushqoqlik ko'ffitsiyenti η orqali ifodalang. Tezlik kichik deb hisoblashingiz mumkin, shunda Stoks qonuni o'rinli bo'ladi va harorat muvozanatga kelganda radiusning o'zgarishini hisobga olmang. Stoks qonunidan kelib chiqadigan qarshilik kuchi $F = 6\pi\eta R_0 u$ ga teng. (1,6 ball)

2.5 $\eta = 1,8 \times 10^{-5} \text{ kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$ qiymatdan foydalanib, u ning son qiymatini hisoblang. (0,4 ball)

Yuqoridagi hisob-kitoblar shuni ko'rsatadiki, sirt tarangligi γ ishtirok etgan hadlar nati-janing aniqligiga juda kam ta'sir qiladi. Quyidagi barcha savollarda sirt taranglik hadlarini hisobga olmasangiz ham bo'ladi.

*Nazariy tur: 12-iyul 2011-yil
2-masala, 2-sahifa (jami 2 tadan)*

2.6 Agar ushbu sferik pufakcha endi umumiy q zaryadi bilan tekis zaryadlangan bo'lsa, yangi R_1 radiusni R_0, P_a, q va erkin fazoning elektr doimiysi (dielektrik singdiruvchanligi) ε_0 orqali tasvirlovchi tenglamani toping. (2,0 ball)

2.7 Umumiy zaryad juda katta emas deb hisoblang (ya'ni, $\frac{q^2}{\varepsilon_0 R_0^4} \ll P_a$) va pufakcha faqat o'z radiusida kichik o'sishga ega bo'ladi, $R_1 = R_0 + \Delta R$ deb olib, ΔR ni toping. Kichik $x \ll 1$ qiymatlar uchun $(1+x)^n \approx 1+nx$ berilgan. (0,7 ball)

2.8 Pufakcha harakatsiz havoda harakatsiz muallaq turishi uchun ushbu q zaryadning moduli $t, \rho_a, \rho_s, \varepsilon_0, R_0, P_a$ orqali nimaga teng bo'lishi kerak? Shuningdek, q ning son qiymatini hisoblang. Erkin fazoning elektr doimiysi $\varepsilon_0 = 8,85 \times 10^{-12} \text{ farad/m}$. (1,2 ball)

*Nazariy tur: 12-iyul 2011-yil
3-masala, 1-sahifa (jami 2 tadan)*

3. Rezerford atom yadrosining yuz yilligini xotirlash: Neytral atom tomonidan ionning sochilishi

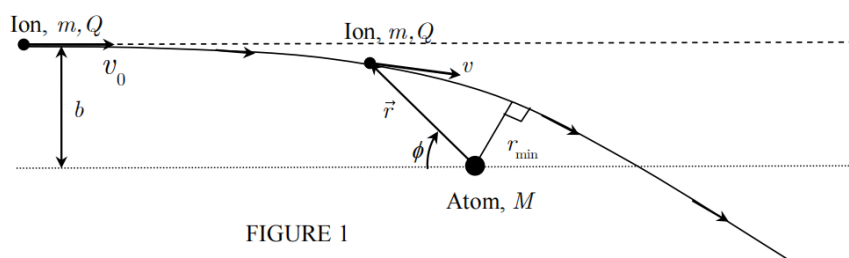


FIGURE 1

Figure 4: 1-rasm

Massasi m , zaryadi Q bo'lgan ion uzoq masofadan boshlang'ich relativistik bo'lmagan v_0 tezlik bilan massasi $M \gg m$ bo'lgan va elektr qutblanuvchanligi α bo'lgan neytral atomning yaqiniga qarab harakatlanmoqda. Nishonni mo'ljalga olish parametri 1-rasmda ko'rsatilganidek b ga teng.

Atom kelayotgan (yaqinlashayotgan) ionning elektr maydoni $\vec{E}$ tomonidan bir zumda qutblanadi. Atomning hosil bo'lgan elektr dipol momenti $\vec{p} = \alpha \vec{E}$ ga teng. Ushbu masalada har qanday nurlanish yo'qotishlarini hisobga olmag.

3.1 2-rasmdagi koordinata boshi O da joylashgan ideal elektr dipol $\vec{p}$ dan r masofada, $\vec{p}$ yo'nalishi bo'ylab elektr maydon kuchlanganligi $\vec{E}_p$ ni hisoblang. (1,2 ball)

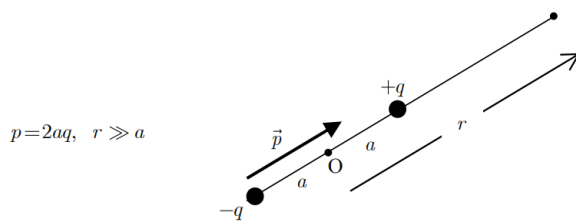


FIGURE 2

Figure 5: 2-rasm

*Nazariy tur: 12-iyul 2011-yil
3-masala, 2-sahifa (jami 2 tadan)*

3.2 Qutblangan atom tufayli ionga ta'sir qiluvchi $\vec{f}$ kuch uchun ifodani toping. Ushbu kuch ion zaryadining ishorasidan qat'i nazar jozibador (tortishish kuchi) ekanligini ko'rsating. (3,0 ball)

3.3 Ion-atom o'zaro ta'sirining elektr potensial energiyasi α, Q va r orqali nimaga teng bo'ladi? (0,9 ball)

3.4 1-rasmda ko'rsatilganidek, eng yaqin kelish masofasi $r_{\min}$ uchun ifodani toping. (2,4 ball)

3.5 Agar nishonni mo'ljalga olish parametri b kritik b_0 qiymatidan kichik bo'lsa, ion atom tomon spiral bo'ylab tushadi. Bunday holda, ion neytrallanadi va atom o'z navbatida zaryadlanadi. Ushbu jarayon "zaryad almashinuvi" o'zaro ta'siri sifatida tanilgan. Ion tomonidan ko'rilgan atomning "zaryad almashinuvi" to'qnashuvi ko'ndalang kesim yuzasi $A = \pi b_0^2$ nimaga teng? (2,5 ball)

*Eksperimental tur: 14-iyul 2011-yil
1-masala, 1-sahifa (jami 4 tadan)*

1. Elektr qora quti: Sig'imli siljish datchigi (sensori)

Tebranish chastotasi f bo'lgan relaksatsion osseptorning tarkibiy qismi bo'lgan sig'imi C bo'lgan kondensator uchun f va C orasidagi bog'liqlik quyidagicha:

$$f = \frac{\alpha}{C + C_s}$$

bu yerda α — doimiy va C_s — bizning zanjirlarning parazit sig'imidir. f chastotani raqamli chastota o'lchagich yordamida kuzatish mumkin.

Ushbu tajribada berilgan elektr qora quti parallel plastinkali kondensatordir. Har bir plastinka bir xil geometrik shakldagi bir nechta kichik tishlardan iborat. C ning qiymatini yuqori plastinkani pastki plastinkaga nisbatan gorizontaal ravishda siljitish orqali o'zgartirish mumkin. Ikki plastinka orasida dielektrik material varag'i mavjud.

Uskunalar: relaksatsion osseptor, relaksatsion osseptorning chastotasini o'lchash uchun raqamli multimetr, ma'lum sig'imli kondensatorlar to'plami, elektr qora quti va batareya.

Ogohlantirish: Batareya kuchlanishini tekshiring va agar kuchlanish 9 V dan kam bo'lsa, yangisini so'rang. Uni yoqishni unutmang.

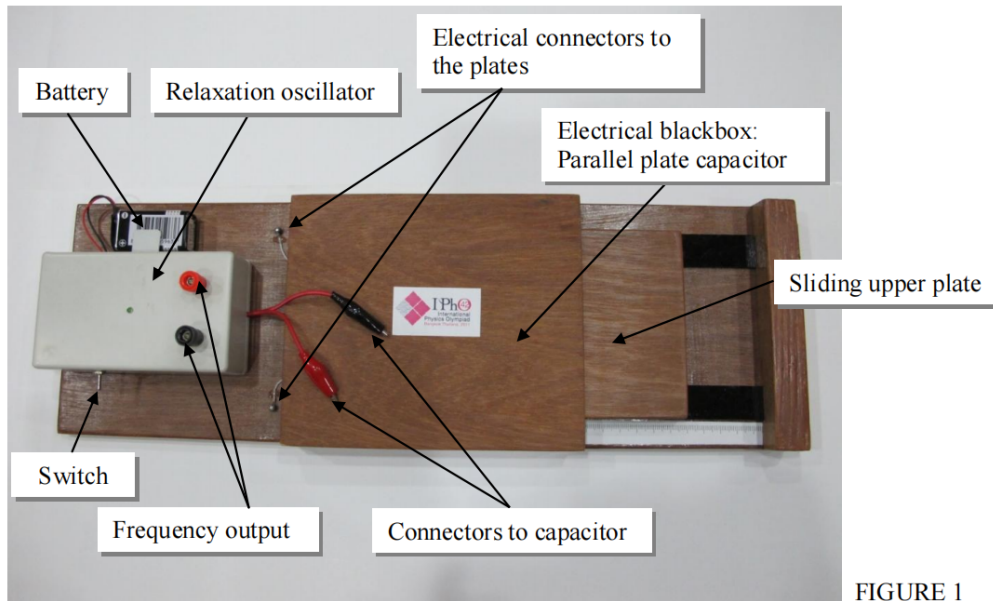


FIGURE 1

Figure 6: 1-rasm

*Eksperimental tur: 14-iyul 2011-yil
1-masala, 2-sahifa (jami 4 tadan)*

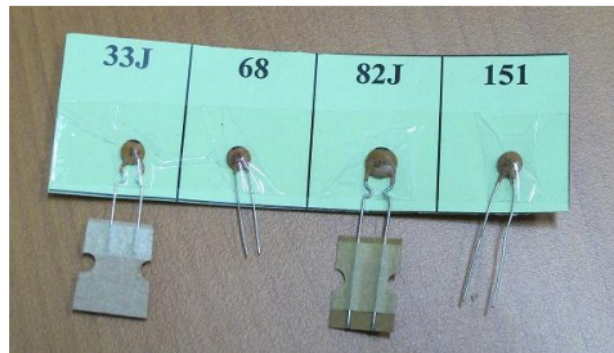


FIGURE 2 Capacitors

Figure 7: 2-rasm: Kondensatorlar

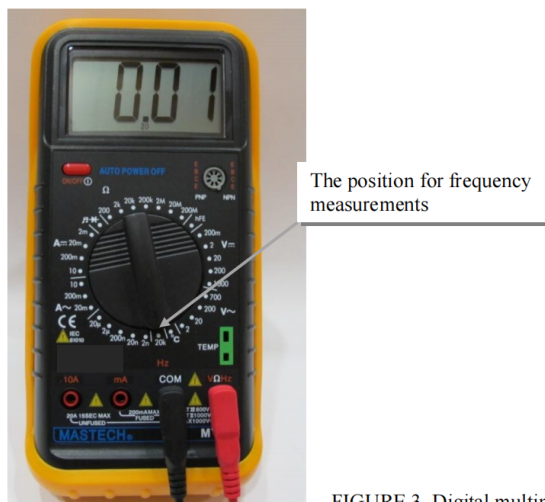


FIGURE 3 Digital multimeter for measuring frequency

Figure 8: 3-rasm: Chastotani o'lchash uchun raqamli multimetr

Table 1: 1-jadval: Nominal sig'im qiymatlari

Kod	Sig'im qiymati (pF)
33J	34 ± 1
68	68 ± 1
82J	84 ± 1
151	150 ± 1

*Eksperimental tur: 14-iyul 2011-yil
1-masala, 3-sahifa (jami 4 tadan)*

Part 1. Kalibrlash (Graduslash)

Ma'lum sig'imli kondensatorlardan foydalangan holda f chastotani o'lchang. α va C_s qiymatlarini topish uchun mos grafik chizing. Xatolik tahlili talab qilinmaydi. (3,0 ball)

Part 2. Parallel plastinkali kondensatorning geometrik shaklini aniqlash

Parallel plastinkali kondensator uchun uchta mumkin bo'lgan geometrik shakl I shakl, II shakl va III shakl quyidagicha berilgan: (6,0 ball)

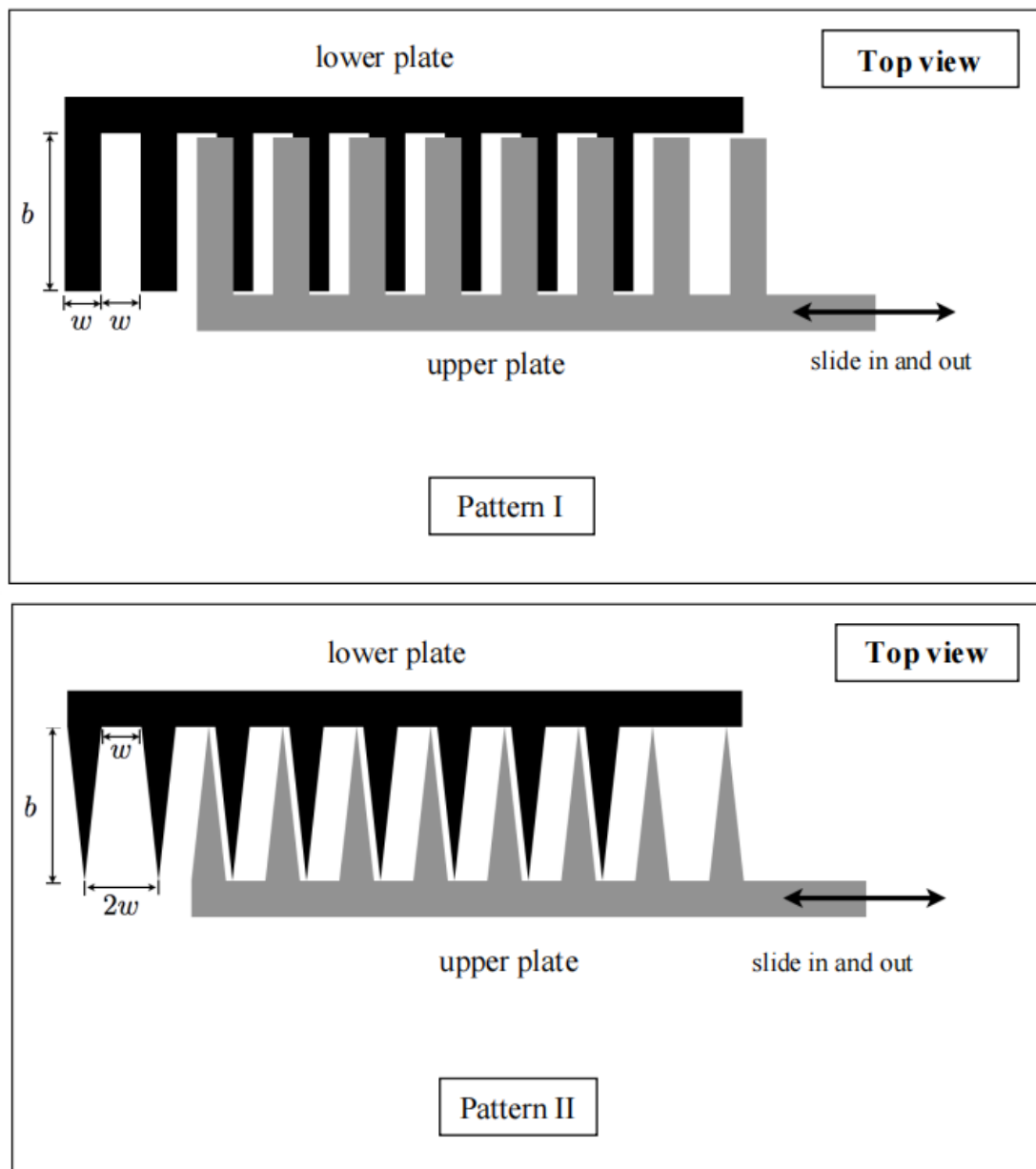


Figure 9: I shakl (Pattern I), II shakl (Pattern II)

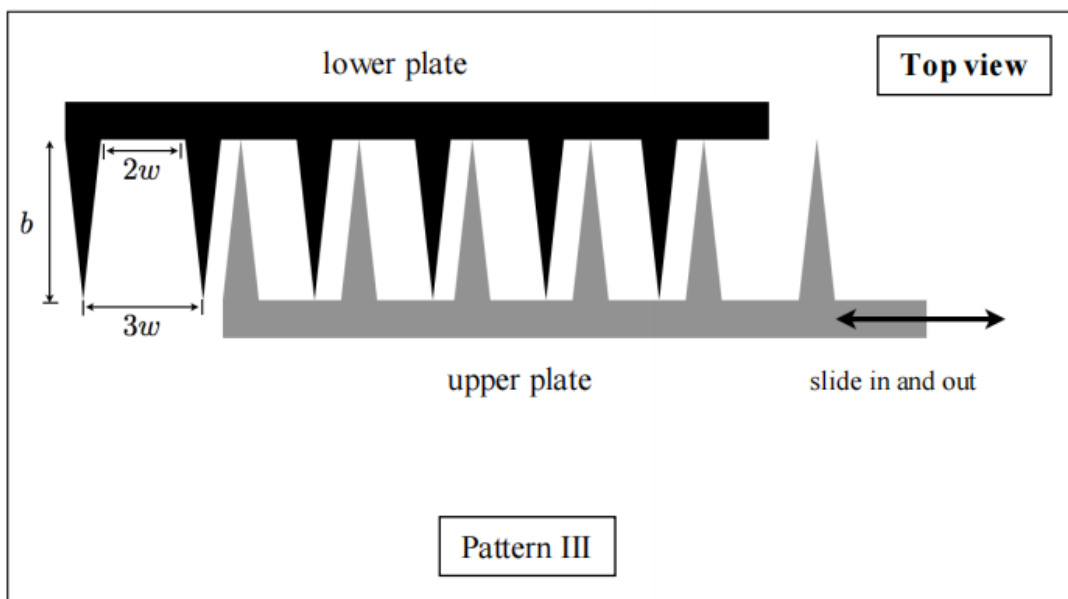


Figure 10: III shakl (Pattern III)

Har bir konfiguratsiya (pattern) uchun yuqori plastinkaning joylashuviga bog'liq holda sig'im C ning o'zgarish grafiklarini sifat jihatidan chizing va gorizonttal o'qni (x -o'qi) belgilang. So'ngra, yuqori plastinkaning siljishiga mos ravishda chastota f ning qiymatlarini o'lchang. Ushbu grafiklardan foydalanib, parallel plastinkali kondensatorning tuzilish shaklini hamda uning o'lchamlarini (b va w ning qiymatlarini) aniqlang. Plastinkalar orasidagi masofa $d = 0,20$ mm. Plastinkalar orasidagi dielektrik qatlamning dielektrik singdiruvchanligi $K = 1,5$. Erkin fazoning dielektrik doimiysi $\varepsilon_0 = 8,85 \times 10^{-12} \text{ F} \cdot \text{m}^{-1}$. Xatoliklarni hisoblash talab etilmaydi.

3-Qism. Raqamli shtangensirkulning aniqlik darajasi (Resolution) [1.0 ball]

Parallel plastinkalarning o'zaro vaziyati o'zgarganda, tizimning sig'imi ma'lum bir qonuniyat asosida o'zgaradi. Ushbu qurilmadan uzunlikni o'lchash uchun raqamli shtangensirkul (digital calipers) sifatida foydalanish mumkin. Agar tajribadagi parallel plastinkali kondensatordan raqamli shtangensirkul sifatida foydalanilsa, 2-qismdagi eksperimental ma'lumotlarga tayanib, uning aniqlik darajasini baholang: ya'ni chastotaning $f \approx 5$ kHz qiymatida o'lchash mumkin bo'lgan eng kichik masofani aniqlang. Yakuniy javob uchun xatolikni baholash talab qilinmaydi.

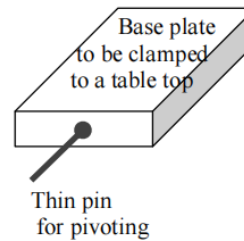
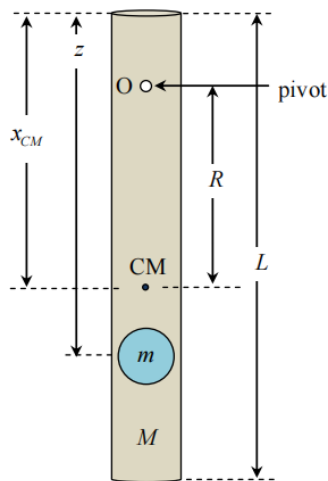
2. Mexanik qora quti (Mechanical Blackbox): ichiga shar joylashtirilgan tsilindr

Massasi m bo'lgan kichik og'ir zarracha (sharcha) massasi M va uzunligi L bo'lgan ichi bo'sh uzun tsilindrning yuqori chetidan z masofada ichki qismga mahkamlangan. Tsilindrning markaziy o'qiga perpendikulyar ravishda bir nechta teshiklar ochilgan. Ushbu teshiklar tsilindrni vertikal tekislikda erkin tebranishini ta'minlash uchun xizmat qiladi (tayanch o'qi vazifasini bajaradi).

Talabalar quyidagi kattaliklarning qiymatlarini hamda ularning xatoliklarini aniqlash uchun zarur bo'lgan mexanik chizg'ichli va vizual (nondestructive) o'lchovlarni bajarishlari lozim:

- i. Ichiga shar o'rnatilgan tsilindrning og'irlik markazi (centre of mass) joylashgan vaziyati. Shuningdek, og'irlik markazini aniqlash uchun mo'ljallangan eksperimental qurilmaning sxematik chizmasini taqdim eting. **[1.0 ball]**
- ii. z masofasi. **[3.5 ball]**
- iii. $\frac{M}{m}$ nisbati. **[3.5 ball]**
- iv. Erkin tushish tezlanishi, g . **[2.0 ball]**

Jihozlar: ichiga shar o'rnatilgan teshikli tsilindr, ingichka mix o'rnatilgan asosiy plastinka, mix qopqog'i, chizg'ich, soniyachalar (stopwatch), ip, qalam va yopishqoq lenta.



x_{CM} is the distance from the top of the cylinder to the centre of mass.
 R is the distance from the pivoting point to the centre of mass.

Figure 11: .

x_{CM} : tsilindrning yuqori chetidan og'irlik markazigacha bo'lgan masofa.

R : tayanch o'qidan (pivot) og'irlik markazigacha bo'lgan masofa.

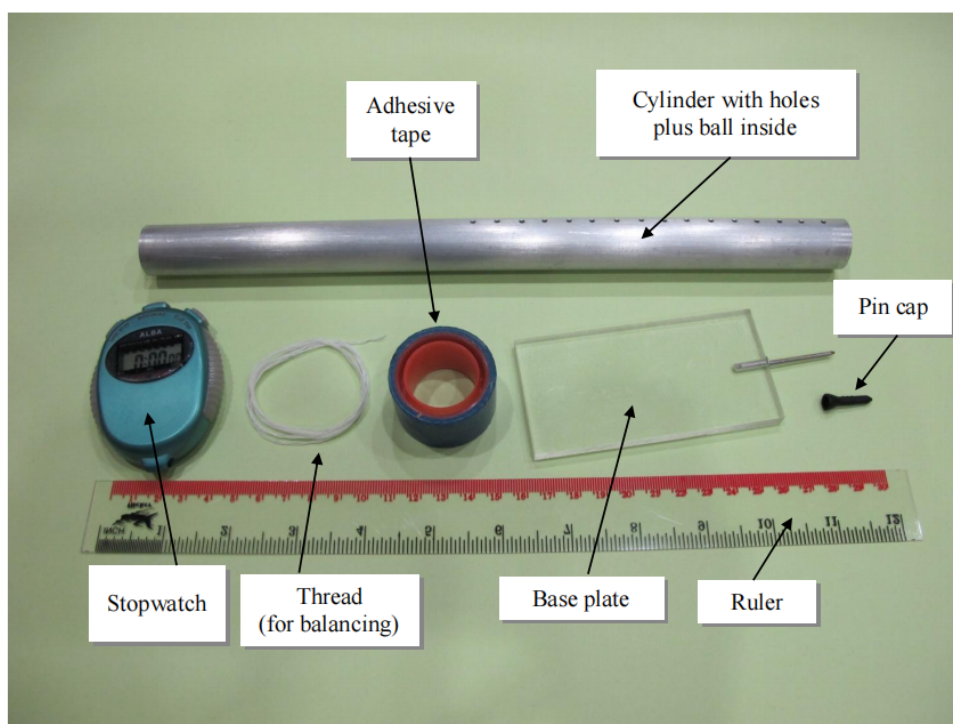


Figure 12: .

[Eksperimental laboratoriya jihozlarining umumiy ko'rinishi]

Rasmda ko'rsatilgan jihozlar: Soniyachalar (Stopwatch), Balanslash uchun ip (Thread), Asosiy plastinka (Base plate), Chizg'ich (Ruler), Yopishqoq lenta (Adhesive tape), Teshikli tsilindr va ichidagi shar, Mix qopqog'i (Pin cap).

Diqqat: Ingichka mix juda o'tkir. Shikastlanishning oldini olish maqsadida, undan foydalanmayotgan vaqtda xavfsizlik uchun unga plastik qopqoqni taqib qo'ying.

Foydali ma'lumotlar:

1. Ushbu fizik mayatnik uchun kichik og'ishlar dGarmonik tebranish tenglamasi quyidagi ko'rinishga ega:

$$(M + m)R^2 + I_{CM} \frac{d^2\theta}{dt^2} \approx -g(M + m)R\theta$$

bu yerda I_{CM} — ichiga shar o'rnatilgan tsilindrning og'irlik markaziga nisbatan inersiya momenti, θ esa burchakli ko'chishdir.

2. Uzunligi L va massasi M bo'lgan tekis, ichi bo'sh tsilindrning markazidan o'tuvchi va uning bo'ylama o'qiga perpendikulyar bo'lgan o'qqa nisbatan inersiya momenti quyidagicha baholanishi mumkin:

$$I \approx \frac{1}{3}M \left(\frac{L}{2} \right)^2$$

3. parallel o'qlar haqidagi Shtayner teoremasi: $I = I_{\text{centre of mass}} + \mathfrak{M}x^2$, bu yerda x — aylanish o'qidan og'irlik markazigacha bo'lgan masofa, va $\mathfrak{M}$ — tizimning to'liq massasi.

4. Ichki qismdagi shar nuqtaviy massa deb qaraladi va u tsilindrning markaziy o'qida joylashgan.
5. Tsilindr devorlari bir jinsli va uning ikki uchidagi qopqoqlarning massasi hisobga olinmaydigan darajada kichik deb hisoblang.

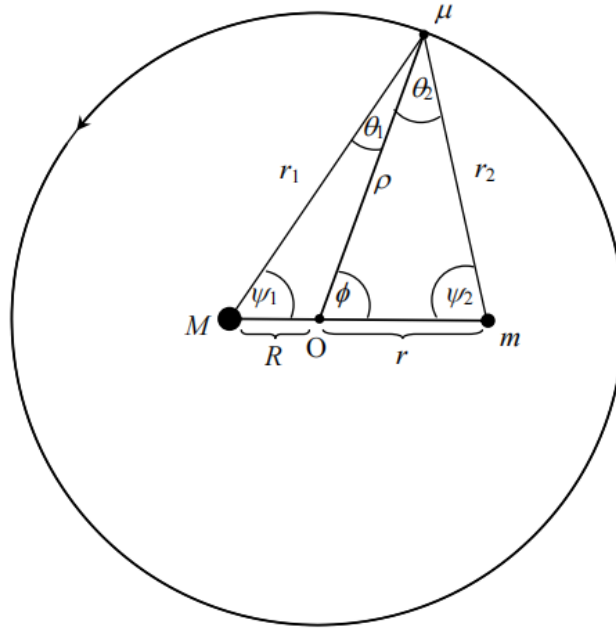


Figure 13: .

I. Yechim

1.1 O nuqta tizimning og'irlik markazi bo'lsin. Shunga ko'ra:

$$MR - mr = 0 \quad (1)$$

$$m\omega_0^2 r = \frac{GMm}{(R+r)^2} \quad (2)$$

$$M\omega_0^2 R = \frac{GMm}{(R+r)^2} \quad (3)$$

(2) tenglamadan yoki keltirilgan massadan (reduced mass) foydalansak, $\omega_0^2 = \frac{G(M+m)}{(R+r)^3}$ kelib chiqadi. Shuning uchun:

$$\omega_0^2 = \frac{G(M+m)}{(R+r)^3} = \frac{GM}{r(R+r)^2} = \frac{Gm}{R(R+r)^2} \quad (4)$$

1.2 μ cheksiz kichik massa bo'lganligi sababli, u na M va na m jismning harakatiga gravitatsion ta'sir ko'rsatmaydi. μ zarrachasi ham M , ham m ga nisbatan qo'zg'almas holatda qolishi uchun quyidagi shart bajarilishi lozim:

$$\frac{GM\mu}{r_1^2} \cos \theta_1 + \frac{Gm\mu}{r_2^2} \cos \theta_2 = \mu\omega_0^2 \rho = \frac{G(M+m)\mu}{(R+r)^3} \rho \quad (5)$$

$$\frac{GM\mu}{r_1^2} \sin \theta_1 = \frac{Gm\mu}{r_2^2} \sin \theta_2 \quad (6)$$

(5) tenglamadan $\frac{GM}{r_1^2}$ ni topib, (4) tenglamaga qo'ysak va $\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2 = \sin(\theta_1 + \theta_2)$ trigonometrik ayniyatidan foydalansak, quyidagiga ega bo'lamiz:

$$m \frac{\sin(\theta_1 + \theta_2)}{r_2^2} = \frac{(M + m)}{(R + r)^3} \rho \sin \theta_1 \quad (7)$$

r_2 va ρ masofalari hamda θ_1 va θ_2 burchaklari o'zaro Sinuslar teoremasi orqali bog'langan:

$$\frac{\sin \psi_1}{\rho} = \frac{\sin \theta_1}{R}, \quad \text{va} \quad \frac{\sin \psi_1}{r_2} = \frac{\sin(\theta_1 + \theta_2)}{R + r} \quad (8)$$

(7) ifodani (6) tenglamaga keltirib qo'ysak:

$$\frac{1}{r_2^3} = \frac{R}{(R + r)^4} \frac{(M + m)}{m} \quad (9)$$

$\frac{m}{M + m} = \frac{R}{R + r}$ ekanligidan, (10) tenglama bizga quyidagicha natija beradi:

$$r_2 = R + r \quad (10)$$

Xuddi shu usulda $\frac{Gm}{r_2^2}$ ni (5) dan topib, (4) ga qo'yish orqali quyidagiga erishamiz:

$$r_1 = R + r \quad (11)$$

Muqobil usul (Alternative):

$$\frac{r_1}{\sin(180^\circ - \phi)} = \frac{R}{\sin \theta_1} \quad \text{va} \quad \frac{r_2}{\sin \phi} = \frac{r}{\sin \theta_2}$$

$$\text{Bundan: } \frac{\sin \theta_1}{\sin \theta_2} = \frac{R}{r} \times \frac{r_2}{r_1} = \frac{m}{M} \times \frac{r_2}{r_1}$$

Ushbu natijani (5) tenglama bilan birlashtirsak, to'g'ridan-to'g'ri $r_1 = r_2$ kelib chiqadi.

Demak, bu uchburchak tomonlari teng bo'lgan muntazam uchburchakdir:

$$\psi_1 = 60^\circ, \quad \psi_2 = 60^\circ \quad (12)$$

ρ masofasi Kosinuslar teoremasiga ko'ra hisoblanadi:

$$\rho^2 = r^2 + (R + r)^2 - 2r(R + r) \cos 60^\circ \quad (13)$$

$$\rho = \sqrt{r^2 + rR + R^2} \quad (14)$$

1.2 qism uchun muqobil yechim (Alternative Solution)

μ cheksiz kichik massa bo'lganligi sababli, u tizim koordinatalariga ta'sir etmaydi. μ muvozanatda qolishi uchun:

$$\frac{GM\mu}{r_1^2} \cos \theta_1 + \frac{Gm\mu}{r_2^2} \cos \theta_2 = \mu\omega_0^2 \rho = \frac{G(M+m)\mu}{(R+r)^3} \rho \quad (4)$$

$$\frac{GM\mu}{r_1^2} \sin \theta_1 = \frac{Gm\mu}{r_2^2} \sin \theta_2 \quad (5)$$

Chizmadan ma'lumki:

$$\frac{r_1}{\sin(180^\circ - \phi)} = \frac{R}{\sin \theta_1} \implies \frac{\sin \theta_1}{\sin \theta_2} = \frac{m}{M} \frac{r_2}{r_1} \quad (6)$$

(5) va (6) tenglamalarni birlashtirsak:

$$r_1 = r_2 \quad (7)$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{m}{M} \quad (8)$$

$$\psi_1 = \psi_2 \quad (9)$$

U holda (4) tenglama quyidagi ko'rinishga keladi:

$$M \cos \theta_1 + m \cos \theta_2 = \frac{(M+m)}{(R+r)^3} r_1^2 \rho \quad (10)$$

(8) va (10) tenglamalardan:

$$\sin(\theta_1 + \theta_2) = \frac{M+m}{M} \frac{r_1^2 \rho}{(R+r)^3} \sin \theta_2 \quad (11)$$

Rasmdan ko'rinib turibdiki,

$$\frac{\rho}{\sin \psi_2} = \frac{r}{\sin \theta_2} \quad (12)$$

(11) va (12) tenglamalardan:

$$\sin(\theta_1 + \theta_2) = \frac{M+m}{M} \frac{r_1^2 r}{(R+r)^3} \sin \psi_2 \quad (13)$$

Shuningdek, rasmdan,

$$(R+r)^2 = r_2^2 - 2r_1 r_2 \cos(\theta_1 + \theta_2) + r_1^2 = 2r_1^2 [1 - \cos(\theta_1 + \theta_2)] \quad (14)$$

(13) va (14) tenglamalardan:

$$\sin(\theta_1 + \theta_2) = \frac{\sin \psi_2}{2[1 - \cos(\theta_1 + \theta_2)]} \quad (15)$$

$$\theta_1 + \theta_2 = 180^\circ - \psi_1 - \psi_2 = 180^\circ - 2\psi_2 \quad (\text{rasmga qarang})$$

$$\therefore \cos \psi_2 = \frac{1}{2}, \quad \psi_2 = 60^\circ, \quad \psi_1 = 60^\circ$$

Demak, M va m tomonlari $(R + r)$ bo'lgan teng tomonli uchburchak hosil qiladi.

μ dan M gacha masofa $R + r$ ga teng.

μ dan m gacha masofa $R + r$ ga teng.

μ dan O gacha masofa

$$\rho = \sqrt{\left(\frac{R+r}{2} - R\right)^2 + \left((R+r)\frac{\sqrt{3}}{2}\right)^2} = \sqrt{R^2 + Rr + r^2}$$

1.3 μ massaning to'liq energiyasi quyidagi ifoda orqali aniqlanadi:

$$E = -\frac{GM\mu}{r_1} - \frac{Gm\mu}{r_2} + \frac{1}{2}\mu \left[\left(\frac{d\rho}{dt}\right)^2 + \rho^2 \dot{\omega}^2 \right] \quad (15)$$

G'alayonlanish (perturbation) faqat radial yo'nalishda bo'lgani uchun burchak momenti saqlanadi ($r_1 = r_2 = \Re$ va $m = M$ holat uchun):

$$E = -\frac{2GM\mu}{\Re} + \frac{1}{2}\mu \left[\left(\frac{d\rho}{dt}\right)^2 + \frac{\rho_0^4 \omega_0^2}{\rho^2} \right] \quad (16)$$

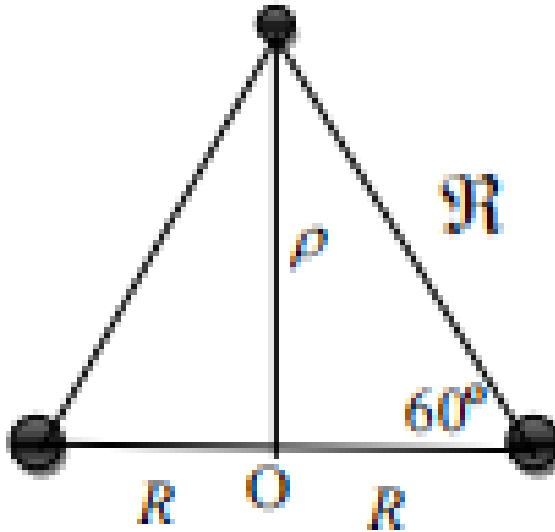


Figure 14: .

Energiyaning saqlanish qonuniga ko'ra,

$$\frac{dE}{dt} = 0$$

$$\frac{dE}{dt} = \frac{2GM\mu}{\mathfrak{R}^2} \frac{d\mathfrak{R}}{dt} + \mu \frac{d\rho}{dt} \frac{d^2\rho}{dt^2} - \mu \frac{\rho_0^4 \omega_0^2}{\rho^3} \frac{d\rho}{dt} = 0 \quad (17)$$

$$\frac{d\mathfrak{R}}{dt} = \frac{d\mathfrak{R}}{d\rho} \frac{d\rho}{dt} = \frac{d\rho}{dt} \frac{rho}{\mathfrak{R}} \quad (18)$$

$$\frac{dE}{dt} = \frac{2GM\mu}{\mathfrak{R}^3} \rho \frac{d\rho}{dt} + \mu \frac{d\rho}{dt} \frac{d^2\rho}{dt^2} - \mu \frac{\rho_0^4 \omega_0^2}{\rho^3} \frac{d\rho}{dt} = 0 \quad (19)$$

$\frac{d\rho}{dt} \neq 0$ bo'lganligi sababli, quyidagiga egamiz:

$$\frac{2GM}{\mathfrak{R}^3} \rho + \frac{d^2\rho}{dt^2} - \frac{\rho_0^4 \omega_0^2}{\rho^3} = 0 \quad \text{yoki} \quad (20)$$

$$\frac{d^2\rho}{dt^2} = -\frac{2GM}{\mathfrak{R}^3} \rho + \frac{\rho_0^4 \omega_0^2}{\rho^3} \quad (21)$$

$\mathfrak{R}_0$ va ρ_0 muvozanat qiymatlaridan kichik chetlanishlar (g'alayonlanishlar) quyidagicha ifodalanadi: $\mathfrak{R} = \mathfrak{R}_0 \left(1 + \frac{\Delta\mathfrak{R}}{\mathfrak{R}_0}\right)$ va $\rho = \rho_0 \left(1 + \frac{\Delta\rho}{\rho_0}\right)$.

U holda:

$$\frac{d^2\rho}{dt^2} = \frac{d^2}{dt^2}(\rho_0 + \Delta\rho) = -\frac{2GM}{\mathfrak{R}_0^3 \left(1 + \frac{\Delta\mathfrak{R}}{\mathfrak{R}_0}\right)^3} \rho_0 \left(1 + \frac{\Delta\rho}{\rho_0}\right) + \frac{\rho_0^4 \omega_0^2}{\rho_0^3 \left(1 + \frac{\Delta\rho}{\rho_0}\right)^3} \quad (21) \quad (22)$$

$(1 + \varepsilon)^n \approx 1 + n\varepsilon$ binomial yoyilmasidan foydalansak:

$$\frac{d^2\Delta\rho}{dt^2} = -\frac{2GM}{\mathfrak{R}_0^3} \rho_0 \left(1 + \frac{\Delta\rho}{\rho_0}\right) \left(1 - \frac{3\Delta\mathfrak{R}}{\mathfrak{R}_0}\right) + \omega_0^2 \rho_0 \left(1 - \frac{3\Delta\rho}{\rho_0}\right) \quad (22) \quad (23)$$

$\Delta\rho = \frac{\mathfrak{R}_0}{\rho_0} \Delta\mathfrak{R}$ munosabatni qo'llab, quyidagini olamiz:

$$\frac{d^2\Delta\rho}{dt^2} = -\frac{2GM}{\mathfrak{R}_0^3} \rho_0 \left(1 + \frac{\Delta\rho}{\rho_0} - \frac{3\rho_0\Delta\rho}{\mathfrak{R}_0^2}\right) + \omega_0^2 \rho_0 \left(1 - \frac{3\Delta\rho}{\rho_0}\right) \quad (23) \quad (24)$$

$\omega_0^2 = \frac{2GM}{\mathfrak{R}_0^3}$ ekanligini hisobga olsak:

$$\frac{d^2\Delta\rho}{dt^2} = -\omega_0^2 \rho_0 \left(1 + \frac{\Delta\rho}{\rho_0} - \frac{3\rho_0\Delta\rho}{\mathfrak{R}_0^2}\right) + \omega_0^2 \rho_0 \left(1 - \frac{3\Delta\rho}{\rho_0}\right) \quad (24) \quad (25)$$

$$\frac{d^2 \Delta \rho}{dt^2} = -\omega_0^2 \rho_0 \left(\frac{4\Delta \rho}{\rho_0} - \frac{3\rho_0 \Delta \rho}{\Re_0^2} \right) \quad (25)$$

$$\frac{d^2 \Delta \rho}{dt^2} = -\omega_0^2 \Delta \rho \left(4 - \frac{3\rho_0^2}{\Re_0^2} \right) \quad (26)$$

Chizmadagi geometrik munosabatga ko'ra, $\rho_0 = \Re_0 \cos 30^\circ$ yoki $\frac{\rho_0^2}{\Re_0^2} = \frac{3}{4}$ ekanligi ma'lum. Buni o'rniga qo'ysak:

$$\frac{d^2 \Delta \rho}{dt^2} = -\omega_0^2 \Delta \rho \left(4 - \frac{9}{4} \right) = -\frac{7}{4} \omega_0^2 \Delta \rho \quad (27)$$

Tebranishning burchak chastotasi: $\frac{\sqrt{7}}{2} \omega_0$.

Muqobil yechim:

$M = m$ bo'lganda $R = r$ va $\omega_0^2 = \frac{G(M+m)}{(R+r)^3} = \frac{GM}{4R^3}$ kelib chiqadi. G'alayonlanmagan holatda μ massaning radial masofasi $\sqrt{3}R$ ga teng. U holda g'alayonlangan radial masofani $\sqrt{3}R + \zeta$ ko'rinishida ifodalash mumkin, bu yerda $\zeta \ll \sqrt{3}R$.

Nyutonning 2-qonuniga ko'ra:

$$-\frac{2GM\mu}{\{R^2 + (\sqrt{3}R + \zeta)^2\}^{3/2}}(\sqrt{3}R + \zeta) = \mu \frac{d^2}{dt^2}(\sqrt{3}R + \zeta) - \mu\omega^2(\sqrt{3}R + \zeta) \quad (1) \quad (29)$$

Burchak momentumining (Harakat miqdori momentining) saqlanish qonunidan:

$$\mu\omega_0(\sqrt{3}R)^2 = \mu\omega(\sqrt{3}R + \zeta)^2 \quad (2) \quad (30)$$

(1) va (2) tenglamalarni birgalikda algebrlik shakldashtirib, $\zeta^2 \approx 0$ ekanligini hisobga olamiz va binomial yoyilmani qo'llaymiz:

$$\begin{aligned} -\frac{2GM}{\{R^2 + (\sqrt{3}R + \zeta)^2\}^{3/2}}(\sqrt{3}R + \zeta) &= \frac{d^2 \zeta}{dt^2} - \frac{\omega_0^2 \sqrt{3}R}{(1 + \zeta/\sqrt{3}R)^3} \\ -\frac{2GM}{\{4R^2 + 2\sqrt{3}\zeta R\}^{3/2}}(\sqrt{3}R + \zeta) &\approx \frac{d^2 \zeta}{dt^2} - \frac{\omega_0^2 \sqrt{3}R}{(1 + \zeta/\sqrt{3}R)^3} \\ -\frac{GM}{4R^3} \sqrt{3}R \frac{(1 + \zeta/\sqrt{3}R)}{(1 + \sqrt{3}\zeta/2R)^{3/2}} &= \frac{d^2 \zeta}{dt^2} - \frac{\omega_0^2 \sqrt{3}R}{(1 + \zeta/\sqrt{3}R)^3} \\ -\omega_0^2 \sqrt{3}R \left(1 - \frac{3\sqrt{3}\zeta}{4R} \right) \left(1 + \frac{\zeta}{\sqrt{3}R} \right) &\approx \frac{d^2 \zeta}{dt^2} - \omega_0^2 \sqrt{3}R \left(1 - \frac{3\zeta}{\sqrt{3}R} \right) \\ \frac{d^2 \zeta}{dt^2} &= -\left(\frac{7}{4} \omega_0^2 \right) \zeta \end{aligned}$$

1.4 Nisbiy tezlik

Aytaylik, v — har bir kosmik kemandi O markazi atrofida aylanma harakatidagi tezligi bo'lsin. Nisbiy tezliklarni A, B va C indekslari orqali belgilaymiz. Masalan, v_{BA} — A kemadagi kuzatuvchiga nisbatan B kemandi tezligi.

Aylanma harakat davri 1 yilga teng: $T = 365 \times 24 \times 60 \times 60$ s (28)

Burchak chastotasi: $\omega = \frac{2\pi}{T}$

Chiziqli tezlik:

$$v = \omega \frac{L}{2 \cos 30^\circ} = 575 \text{ m/s} \quad (29)$$

6

Kemandi tezligi yorug'izlik tezligidan juda kichik $\rightarrow$ Galiley almashtirishlaridan foydalanamiz.

Dekart koordinatalar tizimida O markazga nisbatan B va C kemalarning tezliklari quyidagicha (chizmaga qarang):

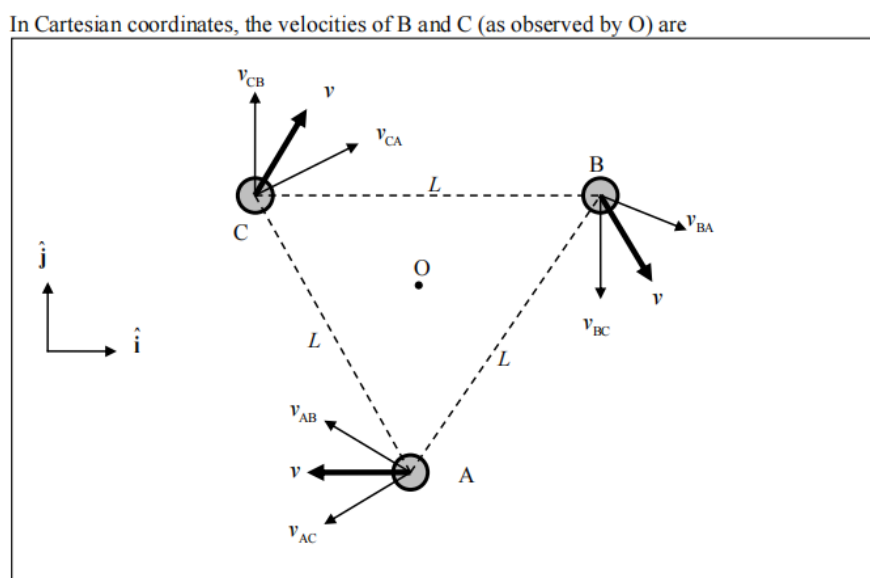


Figure 15: .

B uchun: $\vec{v}_B = v \cos 60^\circ \hat{i} - v \sin 60^\circ \hat{j}$

C uchun: $\vec{v}_C = v \cos 60^\circ \hat{i} + v \sin 60^\circ \hat{j}$

Demak, $\vec{v}_{BC} = -2v \sin 60^\circ \hat{j} = -\sqrt{3}v \hat{j}$

C kemadan kuzatilganda B ning tezligi: $\sqrt{3}v \approx 996 \text{ m/s}$ (30)

E'tibor bering, har bir juftlik uchun nisbiy tezliklar o'zaro anti-parallel (qarama-qarshi) yo'nalgan.

1.4-savol uchun muqobil yechim:

v_{BC} tezlikni kosmik kemalardan birining o'qi atrofida aylanishini hisobga olgan holda ham topish mumkin:

$$v_{BC} = \omega L = \frac{2\pi}{365 \times 24 \times 60 \times 60 \text{ s}} (5 \times 10^6 \text{ km}) \approx 996 \text{ m/s}$$

7

2. YECHIM (SOLUTION)

IPhO 2011, Nazariy musobaqa: Yechim, 2-masala

2.1. Sovun pufagi havo bilan o'ralgan.

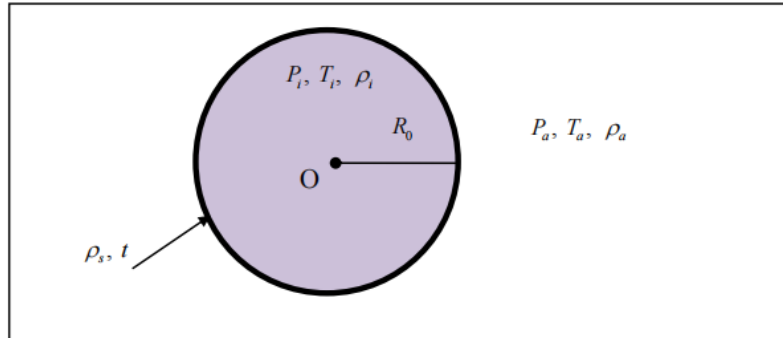


Figure 16: .

Sferani teng ikkiga bo'lik, kuchlarni muvozanatlash (balance) uchun proeksiyalangan yuzadan foydalansak, quyidagiga ega bo'lamiz:

$$\begin{aligned} P_i \pi R_0^2 &= P_a \pi R_0^2 + 2(2\pi R_0 \gamma) \\ P_i &= P_a + \frac{4\gamma}{R_0} \dots (1) \end{aligned}$$

Bosim (pressure) va zichlik (density) ideal gaz qonuni orqali o'zaro bog'langan:

$$PV = nRT \quad \text{yoki} \quad P = \frac{\rho RT}{M}, \quad \text{bu yerda } M = \text{havoning molyar massasi} \dots (2)$$

Ideal gaz qonunini pufak ichidagi va tashqarisidagi havoga qo'llasak, quyidagilarni olamiz:

$$\begin{aligned} \rho_i T_i &= P_i \frac{M}{R} \\ \rho_a T_a &= P_a \frac{M}{R}, \end{aligned}$$

$$\frac{\rho_i T_i}{\rho_a T_a} = \frac{P_i}{P_a} = \left[1 + \frac{4\gamma}{R_0 P_a} \right] \dots (3)$$

2.2. $\gamma = 0,025 \text{ Nm}^{-1}$, $R_0 = 1,0 \text{ cm}$ va $P_a = 1,013 \times 10^5 \text{ Nm}^{-2}$ qiymatlardan foydalanib, nisbatning sonli qiymati:

$$\frac{\rho_i T_i}{\rho_a T_a} = 1 + \frac{4\gamma}{R_0 P_a} = 1 + 0,0001 \dots (4)$$

(Sirt tarangligining (surface tension) ta'siri juda kichik.)

2.3. W = pufakning to'liq og'irligi (weight), F = pufak atrofidagi havo tufayli yuzaga keladigan ko'taruvchi kuch (buoyant force) bo'lsin.

$$\begin{aligned} W &= (\text{parda massasi} + \text{havo massasi})g \\ &= \left(4\pi R_0^2 \rho_s t + \frac{4}{3}\pi R_0^3 \rho_i \right) g \dots (5) \\ &= 4\pi R_0^2 \rho_s t g + \frac{4}{3}\pi R_0^3 \frac{\rho_a T_a}{T_i} \left[1 + \frac{4\gamma}{R_0 P_a} \right] g \end{aligned}$$

Pufak atrofidagi havo tufayli yuzaga keladigan ko'taruvchi kuch:

$$B = \frac{4}{3}\pi R_0^3 \rho_a g \dots (6)$$

Agar pufak tinch havo muhitida suzib yursa (floats):

$$B \geq W$$

$$\frac{4}{3}\pi R_0^3 \rho_a g \geq 4\pi R_0^2 \rho_s t g + \frac{4}{3}\pi R_0^3 \frac{\rho_a T_a}{T_i} \left[1 + \frac{4\gamma}{R_0 P_a} \right] g \dots (7)$$

Qayta tartiblasak, quyidagicha bo'ladi:

$$\begin{aligned} T_i &\geq \frac{R_0 \rho_a T_a}{R_0 \rho_a - 3\rho_s t} \left[1 + \frac{4\gamma}{R_0 P_a} \right] \dots (8) \\ &\geq 307,1 \text{ K} \end{aligned}$$

Ichkaridagi havo taxminan $7,1^\circ\text{C}$ issiqroq bo'lishi kerak.

2.4. Radius o'zgarishini hisobga olmaymiz $\rightarrow$ Radius o'zgarimas qoladi $R_0 = 1,0 \text{ cm}$
(Harorat 307,1 K dan 300 K gacha pasayganda radius aslida 0,8% ga kamayadi. Parda qatlamining o'zi ham biroz qalinlashadi.)

Stokes qonuniga ko'ra qarshilik kuchi (drag force): $F = 6\pi\eta R_0 u$... (9)

Agar pufak yuqoriga yo'nalgan havo oqimida (updraught) suzib yursa:

$$F \geq W - B$$

$$6\pi\eta R_0 u \geq \left(4\pi R_0^2 \rho_s t + \frac{4}{3}\pi R_0^3 \rho_i \right) g - \frac{4}{3}\pi R_0^3 \rho_a g \dots (10)$$

Pufak issiqlik muvozanatida (thermal equilibrium) bo'lganda, $T_i = T_a$:

$$6\pi\eta R_0 u \geq \left(4\pi R_0^2 \rho_s t + \frac{4}{3}\pi R_0^3 \rho_a \left[1 + \frac{4\gamma}{R_0 P_a} \right] \right) g - \frac{4}{3}\pi R_0^3 \rho_a g$$

Qayta tartiblasak, quyidagicha bo'ladi:

$$u \geq \frac{4R_0 \rho_s t g}{6\eta} + \frac{\frac{4}{3}R_0^2 \rho_a g \left(\frac{4\gamma}{R_0 P_a} \right)}{6\eta} \dots (11)$$

2.5. Sonli qiymat: $u \geq 0,36 \text{ m/s}$.

Ikkinchi had (2nd term) birinchi hadga (1st term) qaraganda taxminan 3 tartibga (orders of magnitude) kichik.

Bundan keyin sirt tarangligi hadlarini hisobga olmang.

2.6. Pufak elektrlanganda (electrified), elektr itarish kuchi pufakning kengayishiga va shu orqali ko'taruvchi kuchning ortishiga olib keladi.

Sirtidagi kuch/yuza nisbati: (sirtidagi elektr maydon (e-field)) $\times$ (zaryad/yuza)

Sovun pardasi sirtidagi elektr maydonni hisoblashning ikkita usuli bor.

A. Gauss qonunidan (From Gauss's Law)

Sovun sirtidagi juda yupqa qutichani (pill box) ko'rib chiqing.

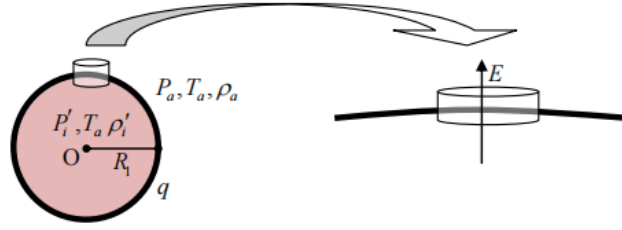


Figure 17: 2-rasm: Yupqa quticha yordamida elektr maydonni aniqlash

E = sovun pardasining o'zidan tashqari barcha boshqa qismlari tomonidan parda sirtida hosil qilingan elektr maydon.

$$\begin{aligned}
 E_q = \text{qutichaning tashqarisidagi to'liq maydon} &= \frac{q}{4\pi\epsilon_0 R_1^2} = \frac{\sigma}{\epsilon_0} \\
 &= E + \text{sirt zaryadi } \sigma \text{ tomondan hosil qilingan elektr ma} \\
 &= E + E_\sigma
 \end{aligned}$$

Qutichaga Gauss qonunini qo'llaganimizda, simmetriyaga ko'ra pardaga perpendikulyar bo'lgan $E_\sigma = \frac{\sigma}{2\epsilon_0}$ ga ega bo'lamiz.

$$\text{Shuning uchun, } E = E_q - E_\sigma = \frac{\sigma}{\epsilon_0} - \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{2\epsilon_0} = \frac{1}{2\epsilon_0} \frac{q}{4\pi R_1^2} \quad \dots (12)$$

B. Bevosita integrallashdan (From direct integration)

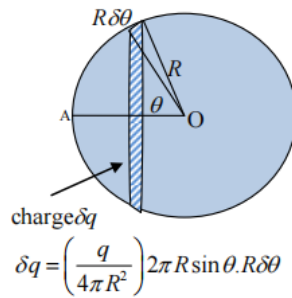


Figure 18: 3-rasm: Sferik sirt bo'ylab elementar zaryad integrali

$$\delta q = \left(\frac{q}{4\pi R^2} \right) 2\pi R \sin \theta \cdot R \delta \theta$$

A nuqtada $\overline{OA}$ yo'nalishidagi elektr maydon intensivligi E ni topish uchun avvalo sirtning o'zida (tashqarisida emas) E_A ni aniqlaymiz:

$$\delta E_A = \frac{1}{4\pi\epsilon_0} \frac{(q/4\pi R_1^2) 2\pi R_1^2 \sin \theta \delta\theta}{\left(2R_1 \sin \frac{\theta}{2}\right)^2} \sin \frac{\theta}{2} = \frac{(q/4\pi R_1^2)}{2\epsilon_0} \cos \frac{\theta}{2} \delta \left(\frac{\theta}{2}\right)$$

$$E_A = \frac{(q/4\pi R_1^2)}{2\epsilon_0} \int_{\theta=0}^{\theta=180^\circ} \cos \frac{\theta}{2} d \left(\frac{\theta}{2}\right) = \frac{(q/4\pi R_1^2)}{2\epsilon_0} \dots (13)$$

Pufak sirtining birlik yuzasiga to'g'ri keladigan itarish kuchi:

$$\left(\frac{q}{4\pi R_1^2}\right) E = \frac{(q/4\pi R_1^2)^2}{2\epsilon_0} \dots (14)$$

P'_i va ρ'_i pufak elektrlangandagi yangi bosim va zichlik bo'lsin.

Bu elektr itarish kuchi gaz bosimi P'_i ni oshiradi. P'_i kattalik dastlabki P_i bilan gaz qonuni orqali quyidagicha bog'langan:

$$P'_i \frac{4}{3} \pi R_1^3 = P_i \frac{4}{3} \pi R_0^3$$

$$P'_i = \left(\frac{R_0}{R_1}\right)^3 P_i = \left(\frac{R_0}{R_1}\right)^3 P_a \dots (15)$$

Oxirgi tenglamada sirt tarangligi hadi hisobga olinmadi.

Yarim sfera proeksiyalangan yuzasidagi kuchlarni muvozanatlaganimizda (yana sirt tarangligi hadini hisobga olmaymiz), quyidagiga ega bo'lamiz:

$$P'_i + \frac{(q/4\pi R_1^2)^2}{2\epsilon_0} = P_a \dots (16)$$

$$P_a \left(\frac{R_0}{R_1}\right)^3 + \frac{(q/4\pi R_1^2)^2}{2\epsilon_0} = P_a$$

Hidlarni qayta guruhlasak, quyidagiga ega bo'lamiz:

$$\left(\frac{R_1}{R_0}\right)^4 - \left(\frac{R_1}{R_0}\right) - \frac{q^2}{32\pi^2\epsilon_0 R_0^4 P_a} = 0 \quad (32)$$

E'tibor bering, kutilganidek $q = 0$ bo'lganda, (17) tenglamadan $\frac{R_1}{R_0} = 1$ kelib chiqadi.

2.7. $\frac{q^2}{32\pi^2\varepsilon_0 R_0^4 P_a} \ll 1$ bo'lgandagi R_1 uchun yaqinlashish yechimi (Approximate solution):

$R_1 = R_0 + \Delta R$ deb olamiz, bu yerda $\Delta R \ll R_0$.

Shuning uchun,

$$\frac{R_1}{R_0} = 1 + \frac{\Delta R}{R_0}, \quad \left(\frac{R_1}{R_0}\right)^4 \approx 1 + 4\frac{\Delta R}{R_0} \quad (33)$$

(17) tenglamaga ko'ra:

$$\Delta R \approx \frac{q^2}{96\pi^2\varepsilon_0 R_0^3 P_a} \quad (34)$$

$$R_1 \approx R_0 + \frac{q^2}{96\pi^2\varepsilon_0 R_0^3 P_a} \approx R_0 \left(1 + \frac{q^2}{96\pi^2\varepsilon_0 R_0^4 P_a}\right) \quad (35)$$

2.8. Pufak quyidagi shart bajarilganda muallaq suzib yuradi (will float if):

$$B \geq W \quad (36)$$

$$\frac{4}{3}\pi R_1^3 \rho_a g \geq 4\pi R_0^2 \rho_s t g + \frac{4}{3}\pi R_0^3 \rho_i g \quad (37)$$

Dastlab, $\gamma \rightarrow 0$ va $R_1 = R_0 \left(1 + \frac{\Delta R}{R_0}\right)$ uchun $T_i = T_a \Rightarrow \rho_i = \rho_a$.

$$\begin{aligned} \frac{4}{3}\pi R_0^3 \left(1 + \frac{\Delta R}{R_0}\right)^3 \rho_a g &\geq 4\pi R_0^2 \rho_s t g + \frac{4}{3}\pi R_0^3 \rho_a g \\ \frac{4}{3}\pi (3\Delta R) \rho_a g &\geq 4\pi R_0^2 \rho_s t g \\ \frac{4}{3}\pi \frac{3q^2}{96\pi^2\varepsilon_0 R_0^3 P_a} \rho_a g &\geq 4\pi R_0^2 \rho_s t g \\ q^2 &\geq \frac{96\pi^2 R_0^5 \rho_s t \varepsilon_0 P_a}{\rho_a} \end{aligned} \quad (38)$$

$$q \approx 256 \times 10^{-9} \text{ C} \approx 256 \text{ nC}$$

Agar sirt tarangligi hadi saqlanib qolsa, u holda quyidagiga ega bo'lamiz:

$$R_1 \approx \left(1 + \frac{q^2/96\pi^2\varepsilon_0 R_0^4 P_a}{\left[1 + \frac{2}{3} \left(\frac{4\gamma}{R_0 P_a}\right)\right]}\right) R_0$$

3-MASALA: YECHIM (QUESTION 3: SOLUTION)

1. Kulon qonunidan (Coulomb's Law) foydalanib, r masofadagi elektr maydonini quyidagicha yozamiz:

$$E_p = \frac{q}{4\pi\epsilon_0(r-a)^2} - \frac{q}{4\pi\epsilon_0(r+a)^2}$$
$$E_p = \frac{q}{4\pi\epsilon_0 r^2} \left(\frac{1}{\left(1 - \frac{a}{r}\right)^2} - \frac{1}{\left(1 + \frac{a}{r}\right)^2} \right) \quad (1)$$

Kichik a qiymatlari uchun binominal yoyilmadan (binomial expansion) foydalanamiz:

$$E_p = \frac{q}{4\pi\epsilon_0 r^2} \left(1 + \frac{2a}{r} - 1 + \frac{2a}{r} \right)$$
$$= + \frac{4qa}{4\pi\epsilon_0 r^3} = + \frac{qa}{\pi\epsilon_0 r^3} \quad (2)$$
$$= \frac{2p}{4\pi\epsilon_0 r^3}$$

2. Ion tomonidan atomda hosil qilingan elektr maydoni:

$$\vec{E}_{\text{ion}} = -\frac{Q}{4\pi\epsilon_0 r^2} \hat{r} \quad (39)$$

U holda induksiyalangan dipol momenti (induced dipole moment) shunchaki:

$$\vec{p} = \alpha \vec{E}_{\text{ion}} = -\frac{\alpha Q}{4\pi\epsilon_0 r^2} \hat{r} \quad (40)$$

(2) tenglamaga ko'ra:

$$\vec{E}_p = \frac{2p}{4\pi\epsilon_0 r^3} \hat{r}$$

Ushbu paytda ion turgan nuqtadagi elektr maydon intensivligi $\vec{E}_p$, (4) tenglamadan foydalangan holda:

$$\vec{E}_p = \frac{1}{4\pi\epsilon_0 r^3} \left[-\frac{2\alpha Q}{4\pi\epsilon_0 r^2} \hat{r} \right] = -\frac{\alpha Q}{8\pi^2 \epsilon_0^2 r^5} \hat{r}$$

ionga ta'sir qilayotgan kuch:

$$\vec{f} = Q \vec{E}_p = -\frac{\alpha Q^2}{8\pi^2 \epsilon_0^2 r^5} \hat{r} \quad (41)$$

"—" isorasi ushbu kuchning tortishish xarakteriga ega ekanligini anglatadi va Q^2 esa kuchning ion zaryadi ishorasidan qat'iy nazar tortishish kuchi bo'lishini bildiradi.

3. Ion-atom tizimining potensial energiyasi quyidagicha aniqlanadi:

$$U = \int_r^{\infty} \vec{f} \cdot d\vec{r} \quad (42)$$

Bundan foydalangan holda:

$$U = \int_r^{\infty} \vec{f} \cdot d\vec{r} = -\frac{\alpha Q^2}{32\pi^2 \varepsilon_0^2 r^4} \quad (43)$$

[Eslatma: Talabalar $-\vec{p} \cdot \vec{E}$ ifodasidan foydalanishlari mumkin, bu faqat oldindagi koeffitsiyentni o'zgartiradi.]

4. $r_{\min}$ nuqtada, Burchak impulsining saqlanish qonuniga (Principle of Conservation of Angular Momentum) binoan:

$$mv_{\max} r_{\min} = mv_0 b$$

$$v_{\max} = v_0 \frac{b}{r_{\min}} \quad (8)$$

Va Energiya saqlanish qonuniga (Principle of Conservation of Energy) muvofiq:

$$\frac{1}{2}mv_{\max}^2 + \frac{-\alpha Q^2}{32\pi^2 \varepsilon_0^2 r_{\min}^4} = \frac{1}{2}mv_0^2 \quad (44)$$

(12) va (13) tenglamalardan:

$$\left(\frac{b}{r_{\min}}\right)^2 - \frac{\alpha Q^2 / \frac{1}{2}mv_0^2}{32\pi^2 \varepsilon_0^2 b^4} \left(\frac{b}{r_{\min}}\right)^4 = 1$$

$$\left(\frac{r_{\min}}{b}\right)^4 - \left(\frac{r_{\min}}{b}\right)^2 + \frac{\alpha Q^2}{16\pi^2 \varepsilon_0^2 mv_0^2 b^4} = 0 \quad (10)$$

(14) tenglamaning ildizlari quyidagicha bo'ladi:

$$r_{\min} = \frac{b}{\sqrt{2}} \left[1 \pm \sqrt{1 - \frac{\alpha Q^2}{4\pi^2 \varepsilon_0^2 mv_0^2 b^4}} \right]^{\frac{1}{2}} \quad (45)$$

[E'tibor bering, (14) tenglama b ning o'zi nolga teng bo'lmasa, $r_{\min}$ nol bo'la olmasligini anglatadi.]

Ushbu ifoda $Q = 0$ bo'lganda ham o'rinli bo'lishi kerak, bundan kelib chiqadiki:

$$r_{\min} = \frac{b}{\sqrt{2}} [1 \pm 1]^{\frac{1}{2}}$$

$r_{\min} = b$ bo'lishi uchun biz "+" ishorasini tanlashimiz kerak.

Shunday qilib,

$$r_{\min} = \frac{b}{\sqrt{2}} \left[1 + \sqrt{1 - \frac{\alpha Q^2}{4\pi^2 \varepsilon_0^2 mv_0^2 b^4}} \right]^{\frac{1}{2}} \quad (46)$$

5. Spiral trayektoriya (16) ildiz ostidagi ifoda mavhum (imaginary) bo'lganda yuz beradi (chunki bu holda eng yaqin masofaga kelish nuqtasi mavjud bo'lmaydi).

$r_{\min}$ haqiqiy son bo'lish sharti:

$$1 \geq \frac{\alpha Q^2}{4\pi^2 \varepsilon_0^2 m v_0^2 b^4}$$

$$b \geq b_0 = \left(\frac{\alpha Q^2}{4\pi^2 \varepsilon_0^2 m v_0^2} \right)^{\frac{1}{4}} \quad (13)$$

$b < b_0 = \left(\frac{\alpha Q^2}{4\pi^2 \varepsilon_0^2 m v_0^2} \right)^{\frac{1}{4}}$ bo'lganda, ion atom bilan to'qnashadi (will collide).

Shunday qilib, ion tomondan qaralganda atom quyidagi ko'ndalang kesim yuzasiga (cross-sectional area) ega bo'ladi A :

$$A = \pi b_0^2 = \pi \left(\frac{\alpha Q^2}{4\pi^2 \varepsilon_0^2 m v_0^2} \right)^{\frac{1}{2}} \quad (47)$$

1-qism. Kalibrlash

f va C orasidagi berilgan munosabatdan:

$$f = \frac{\alpha}{C + C_s} \quad \Leftrightarrow \quad \frac{1}{f} = \frac{1}{\alpha} C + \frac{C_s}{\alpha} \quad (48)$$

Ya'ni, nazariy jihatdan Y o'qidagi $1/f$ ning X o'qidagi C ga grafigi chiziqli bo'lishi kerak, bunda qiyalik $\frac{1}{\alpha}$ va Y o'qidagi kesishma $\frac{C_s}{\alpha}$ ga teng.

Quyidagi jadvalda C (X o'qiga qo'yiladi), f va qo'shimcha ravishda Y o'qiga qo'yiladigan $1/f$ ning o'lchangan qiymatlari keltirilgan.

C (pF)	f (kHz)	$1/f$ (ms)
33	13.94	0.0717
68	8.30	0.1205
82	6.99	0.1431
151	4.17	0.2398
233	2.79	0.3584
219	2.98	0.3356
184	3.48	0.2874
150	4.20	0.2381
115	5.24	0.1908
101	5.89	0.1698

Table 2: Kalibrlash jadvali

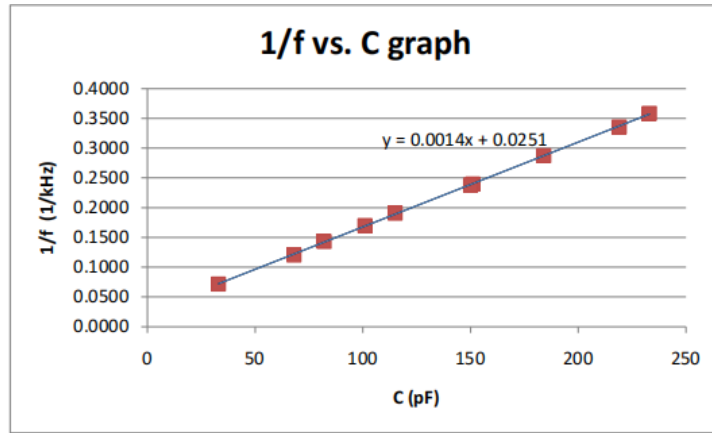


Figure 19: 1-rasm: NAQSh I — C ning vaziyatga bog'liqligining kutilayotgan grafigi

Ushbu grafikdan qiyalik ($\frac{1}{\alpha}$) va Y o'qidagi kesishma ($\frac{C_s}{\alpha}$) mos ravishda 0.0014 s/nF va 0.0251 ms ga teng.

Shunday qilib,

$$\alpha = \frac{1}{\text{qiyalik}} = \frac{1}{0.0014 \text{ s / nF}} = 714 \text{ nF/s} \quad (49)$$

va

$$C_s = \frac{\text{kesishma}}{\text{qiyalik}} = \frac{0.0251 \text{ ms}}{0.0014 \text{ s / nF}} = 17.9 \text{ pF} \quad (50)$$

talab qilinganidek.

2-qism. Parallel plastinkali kondensatorning geometrik shaklini aniqlash

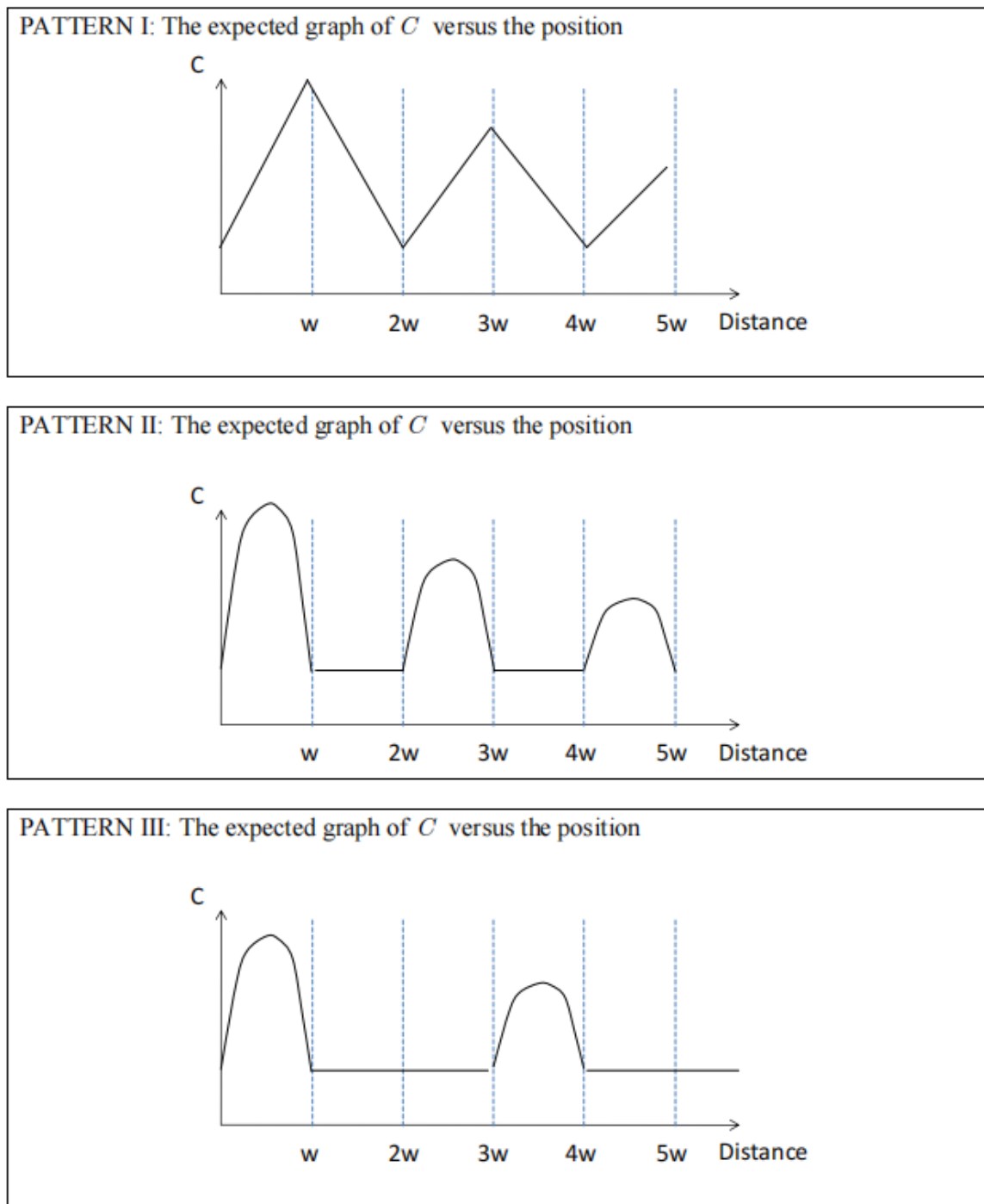


Figure 20: 1-rasm: NAQSh I — C ning vaziyatga bog'liqligining kutilayotgan grafigi, 2-rasm: NAQSh II — C ning vaziyatga bog'liqligining kutilayotgan grafigi, 3-rasm: NAQSh III — C ning vaziyatga bog'liqligining kutilayotgan grafigi

f va C ni x (ikki plastinka orasidagi siljish masofasi) ga nisbatan o'lchash orqali quyidagi ma'lumotlar va grafiklar olingan.

x (mm)	f (kHz)	C (pF)	x (mm)	f (kHz)	C (pF)
0	7.41	77.9	30	4.94	126.1
1	8.09	69.8	31	5.52	110.9
2	8.64	64.2	32	6.19	96.9
3	9.30	58.3	33	6.48	91.7
4	9.30	58.3	34	6.64	89.1
5	8.21	68.5	35	5.72	106.4
6	7.02	83.3	36	5.08	122.1
7	6.40	93.1	37	4.39	144.2
8	5.98	100.9	38	4.06	157.4
9	5.91	102.4	39	3.97	161.4
10	6.38	93.5	40	4.32	146.8
11	6.96	84.1	41	4.86	128.5
12	7.61	75.4	42	5.33	115.5
13	8.40	66.5	43	6.05	99.6
14	8.20	68.6	44	5.98	100.9
15	7.13	81.7	45	5.14	120.5
16	6.37	93.6	46	4.47	141.3
17	5.96	101.3	47	3.93	163.3
18	5.38	114.3	48	3.74	172.5
19	5.33	115.5	49	3.64	177.7
20	5.72	106.4	50	3.93	163.3
21	6.34	94.2	51	4.30	147.6
22	6.85	85.8	52	4.91	127.0
23	7.53	76.4	53	5.46	112.3
24	7.23	80.3	54	5.49	111.6
25	6.33	94.3	55	4.64	135.4
26	5.56	110.0	56	4.07	157.0
27	5.36	114.8	57	3.62	178.8
28	4.73	132.5	58	3.36	194.1
29	4.53	139.2			

Table 3: x , f va C o'lchovlari

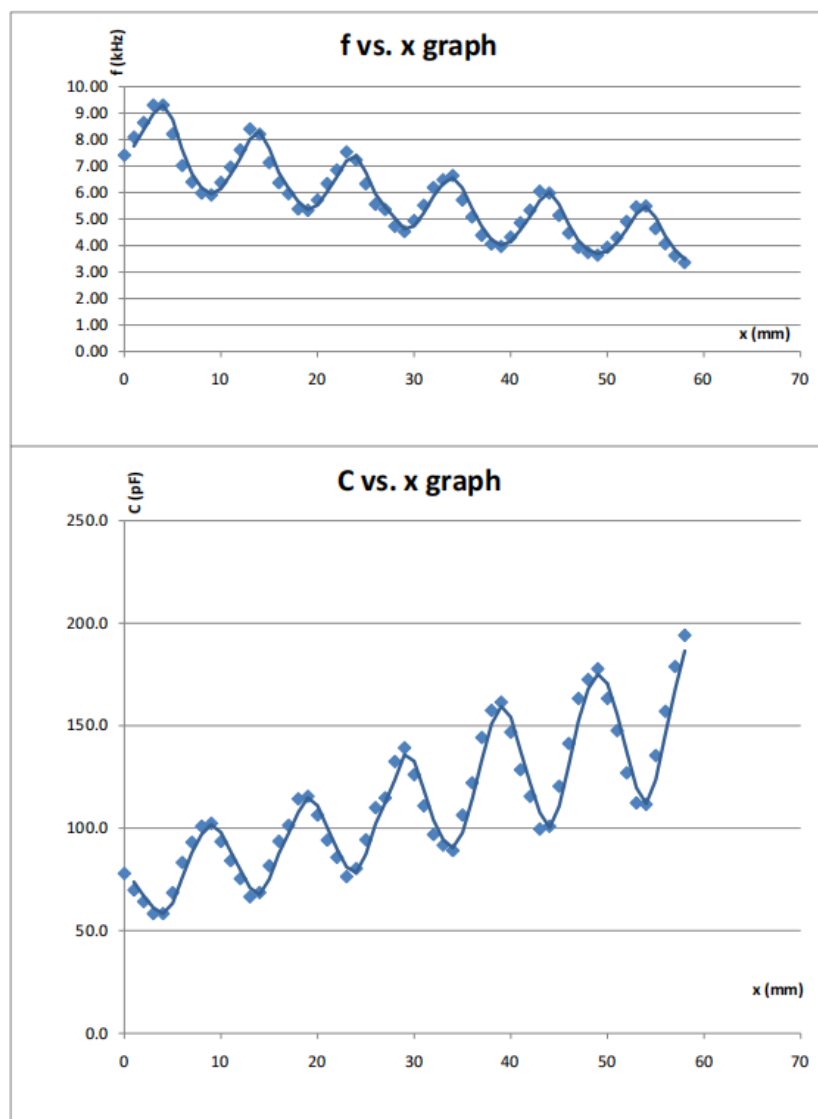


Figure 21: 4-rasm: f ning x ga bog'liqligi grafigi

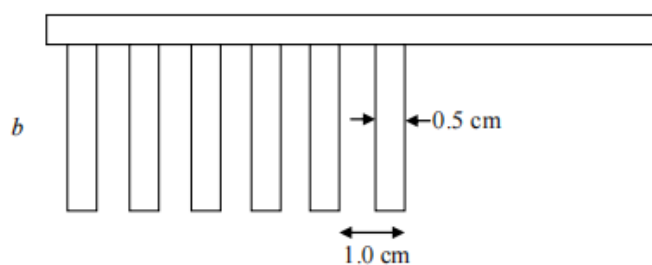


Figure 22: 5-rasm: C ning x ga bog'liqligi grafigi

Grafikning davriyligidan $\text{davr} = 1.0 \text{ sm}$ ga teng.

Oddiy mumkin bo'lgan konfiguratsiya:

C ning x ga grafigidan olingan C qiymatlarining eng katta cho'qqilari quyidagi jadvalda keltirilgan. Ushbu maksimal C lar (Y o'qida) tugunlar (X o'qida) ga nisbatan chizilgan.

Tugun	C_{maks} (pF)
1	105.1
2	118.6
3	139.5
4	163.7
5	182.1

Table 4: Maksimal sig'ímlar

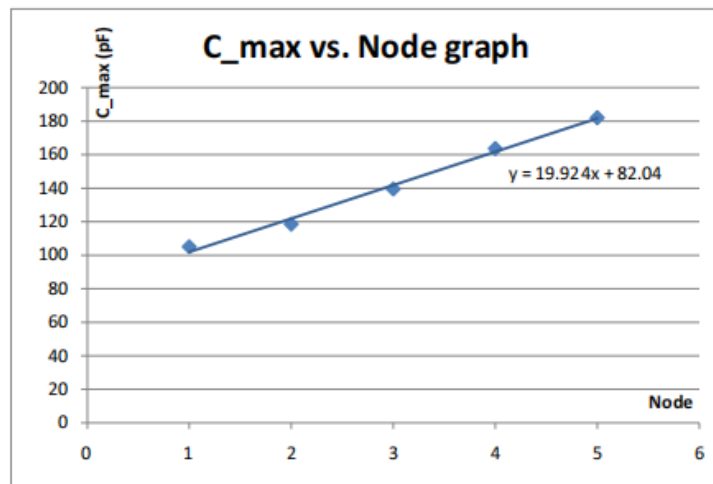


Figure 23: 1-rasm: NAQSh I — C ning vaziyatga bog'liqligining kutilayotgan grafigi

Bu grafik chiziqli bo'lib, uning qiyaligi har bir bo'lak uchun sig'imning pasayishi $\Delta C = 19.9$ pF ga teng.

Plastinkalar orasidagi masofa $d = 0.20$ mm, $K = 1.5$ berilgan holda,

$$\Delta C \approx \frac{K\varepsilon_0 A}{d}, \quad (51)$$

va

$$A = 5 \times 10^{-3} \text{ m} \times b \text{ mm} \times 10^{-3} \text{ m}^2 \quad (52)$$

U holda, $b \text{ mm} \approx \frac{\Delta C \cdot d}{K\varepsilon_0 \times 10^{-3} \times 5 \times 10^{-3}} \approx 60 \text{ mm}$ agar plastinkalar orasidagi muhit dielektrik bo'lib, uning o'tkazuvchanligi $K = 1.5$ ga teng bo'lsa.

III qism. Raqamli mikrometrning aniqlik darajasi (Resolution of digital micrometer)

f va C orasidagi berilgan bog'liqlikdan: $f = \frac{\alpha}{C + C_s},$

$$\begin{aligned}
\Delta f &\approx \left| \frac{df}{dC} \right| \Delta C = \left| \frac{-\alpha}{(C + C_s)^2} \right| \Delta C \\
&= \frac{f^2}{\alpha} \Delta C \\
&\Leftrightarrow \Delta C = \frac{\alpha}{f^2} \Delta f
\end{aligned}$$

Va C kattaligi x ga chiziqli bog'liq bo'lgani uchun: $C = mx + \beta \Rightarrow \Delta C = m\Delta x$.

Bundan kelib chiqadiki,

$$\Delta x = \frac{\alpha}{mf^2} \Delta f \quad (53)$$

bu yerda Δf — multimetr aniqlay oladigan chastotaning eng kichik o'zgarishi, x_0 esa $f = 5 \text{ kHz}$ chastotaga mos keladigan masofa, m esa C ning x ga bog'liqlik grafigidagi $f = 5 \text{ kHz}$ atrofida og'ish burchagi koeffitsiyentidir (gradient).

f ning x ga bog'liqlik grafigidan, $f = 5 \text{ kHz}$ nuqtada, C ning x ga bog'liqlik grafigining og'ish koeffitsiyenti ushbu soha atrofida o'lchanadi.

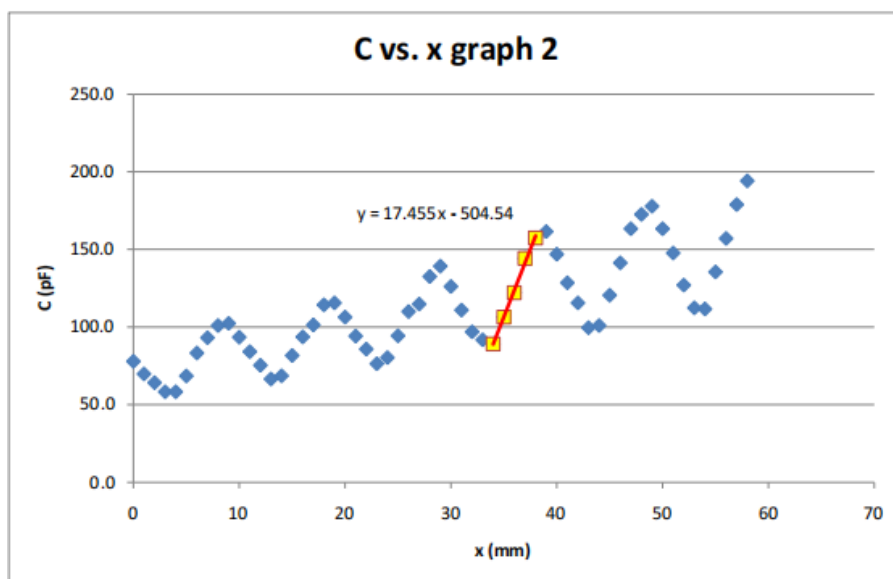


Figure 24: .

C ning x ga bog'liqlik grafigi 2 (C vs. x graph 2)

Ushbu grafikdan chiziqli regression tenglama quyidagicha topiladi:

Grafigidan ko'rinib turibdiki, burchak koeffitsiyenti: $m = 17.5 \text{ pF/mm} = 1.75 \times 10^{-8} \text{ F/m}$.

Ushbu m qiymatidan foydalanib, $f = 5 \text{ kHz}$, $\alpha = 714 \text{ nF/s}$ va $\Delta f = 0.01 \text{ kHz}$ bo'lganda:

$$\Delta x = \frac{714 \times 10^{-9}}{(1.75 \times 10^{-8}) \times (5 \times 10^3)^2} \times (0.01 \times 10^3) = 0.016 \text{ mm} \quad (54)$$

Diqqat: C ning x ga bog'liqlik grafigi chiziqli, chunki C (f emas) x ga to'g'ridan-to'g'ri bog'liqdir.

Aniqlik darajasini topishning muqobil usuli (to'liq aniq bo'lmagan usul):

f ning x ga bog'liqlik grafigi va jadvaldagi $f = 5 \text{ kHz}$ atrofida ma'lumotlardan foydalanilsa, f chastota 1 kHz ga o'zgarganda ($\Delta f = 1 \text{ kHz}$), x masofa taxminan 1.5 mm ga o'zgarishi aniqlanadi ($\Delta x \approx 1.5 \text{ mm}$). Shunga ko'ra, chastota $\Delta f = 0.01 \text{ kHz}$ (eng kichik o'zgarish qadami) bo'lganda, siljigan masofa $\Delta x \approx 0.015 \text{ mm}$ ni tashkil etadi.

2-MASALA YECHIMI: Mexanik qora quti (A cylinder with a ball inside)

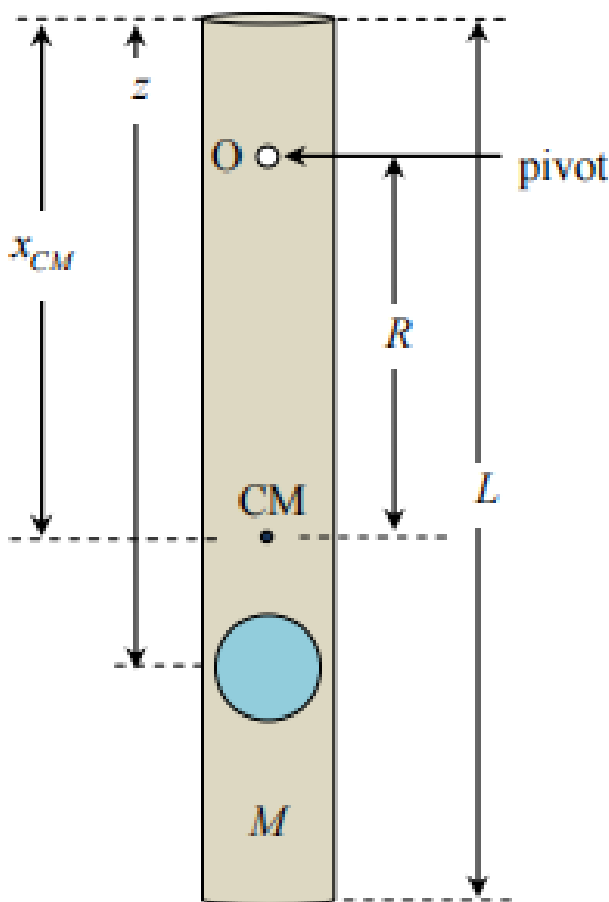


Figure 25: .

i, ii, iii bo'limlaridagi kerakli qiymatlarni hisoblash uchun biz quyidagilarni bilishimiz kerak:

- Naycha va ichidagi zarracha (jism) tizimining og'irlik markazi (centre of mass) joylashuvi, bu z, m, M kattaliklariga bog'liq: x_{CM}
- Yuqoridagi tizimning inersiya momenti (moment of inertia).

Og'irlik markazining (CM) joylashuvi muvozanatlash usuli orqali topilishi mumkin. I_{CM} inersiya momenti esa naycha va jism tizimining tebranish davri (period of oscillation) orqali hisoblanadi.

Grafik chizish uchun parametrlarni tanlashning analitik bosqichlari:

$$I. \quad x_{CM} = \frac{mz + M\frac{L}{2}}{m + M} \quad (55)$$

L uzunligi chizg'ich yordamida oson o'lchanadi. x_{CM} esa naycha va ichki jism tizimini muvozanatlash orqali aniqlanadi.

II. Kichik amplitudali tebranishlar uchun ixtiyoriy O nuqtaga nisbatan tebranish davri T quyidagi harakat tenglamasidan topiladi:

$$\{(M+m)R^2 + I_{CM}\}\ddot{\theta} = -g(M+m)R \sin \theta \approx -g(M+m)R\theta \quad (56)$$

$$T = 2\pi \sqrt{\frac{I_{CM} + (M+m)R^2}{g(M+m)R}} \quad (57)$$

bu yerda,

$$\begin{aligned} I_{CM} &= \frac{1}{3}M \left(\frac{L}{2}\right)^2 + M \left(x_{CM} - \frac{L}{2}\right)^2 + m(z - x_{CM})^2 \\ &= \frac{1}{3}ML^2 + Mx_{CM}^2 - MLx_{CM} + m(z - x_{CM})^2 \end{aligned} \quad (58)$$

E'tibor bering:

$$T^2 \frac{g(M+m)}{4\pi^2} = \frac{I_{CM}}{R} + (M+m)R \quad (59)$$

A usuli: (Chiziqli grafik usuli / linear graph method)

(5) tenglamani quyidagi ko'rinishga keltirish mumkin:

$$T^2 R = \left(\frac{4\pi^2}{g}\right) R^2 + \frac{4\pi^2 I_{CM}}{(M+m)g} \quad (60)$$

Demak, $T^2 R$ ning R^2 ga bog'liqlik grafigi to'g'ri chiziqli bo'ladi va uning parametrlari:

$$\text{Burchak koeffitsiyenti (Slope): } \alpha = \frac{4\pi^2}{g} \quad (61)$$

$$\text{y-o'qini kesib o'tish nuqtasi (y-intercept): } \beta = \frac{4\pi^2 I_{CM}}{(M+m)g} \quad (62)$$

$$\text{Bundan, } I_{CM} = (M+m) \frac{\beta}{\alpha} \quad (63)$$

$$\text{Erkin tushish tezlanishi } g \text{ ning qiymati (7) tenglamadan: } g = \frac{4\pi^2}{\alpha} \quad (64)$$

B usuli: (Minimal nuqta egri chizig'i usuli / minimum point curve method)

(5) tenglamadan kelib chiqadiki, T funksiya quyidagi nuqtada o'zining minimal qiymatiga ega bo'ladi:

$$R = R_{\min} \equiv \sqrt{\frac{I_{CM}}{M + m}} \quad (65)$$

Shunday qilib, $R_{\min}$ qiymatni T ning R ga bog'liqlik grafigidan aniqlash mumkin.

Va demak:

$$I_{CM} = (M + m)R_{\min}^2 \quad (66)$$

Ushbu (12) tenglama (1) tenglama bilan birgalikda bizga noma'lum z va $\frac{M}{m}$ nisbatni hisoblash imkonini beradi.

$R = R_{\min}$ qiymatida (5) tenglama quyidagi ko'rinishga keladi:

$$\begin{aligned} T_{\min}^2 \frac{g(M + m)}{4\pi^2} &= (M + m)R_{\min} + (M + m)R_{\min} \\ g &= \frac{2R_{\min}}{T_{\min}^2} \times 4\pi^2 = \frac{8\pi^2 R_{\min}}{T_{\min}^2} \end{aligned} \quad (67)$$

bu ifoda orqali g ning qiymati osongina hisoblanadi.

O'lchov natijalari

$$L = 30.0 \text{ sm} \pm 0.1 \text{ sm} \quad (68)$$

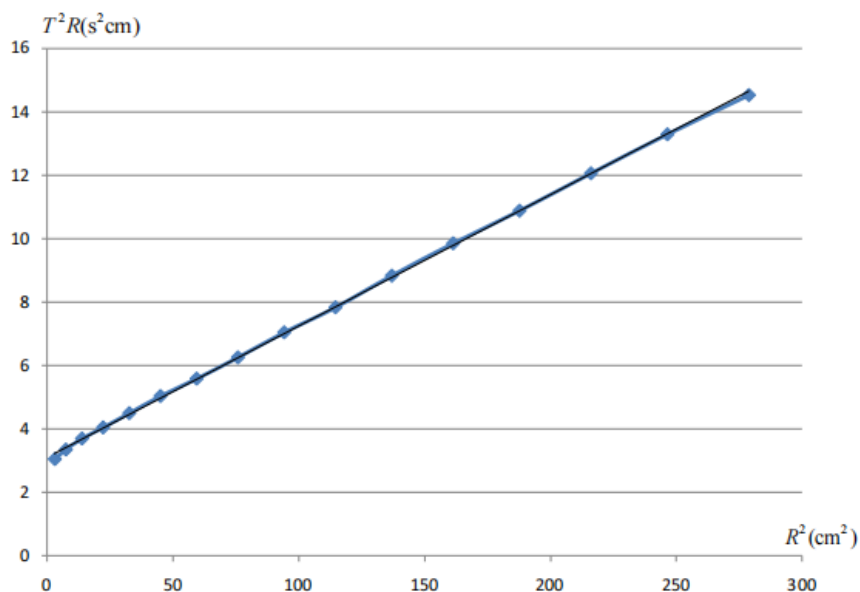
$$x_{CM} = 17.8 \text{ sm} \pm 0.1 \text{ sm} \text{ (yuqoridan)} \quad (69)$$

$x_{CM} - R$ (sm)	20 sikl uchun vaqt (s)	T (s)	R (sm)	R^2 (sm ²)	$T^2 R$ (s ² ·sm)
1.1	18.59 18.78 18.59	0.933	16.7	278.9	14.53
2.1	18.44 18.25 18.53	0.920	15.7	246.5	13.29
3.1	18.10 18.09 18.15	0.906	14.7	216.1	12.06
4.1	17.88 17.78 17.81	0.891	13.7	187.7	10.88
5.1	17.69 17.50 17.65	0.881	12.7	161.3	9.85
6.1	17.47 17.38 17.28	0.869	11.7	136.9	8.83
7.1	17.06 17.06 17.22	0.856	10.7	114.5	7.83
8.1	17.06 17.00 17.06	0.852	9.7	94.1	7.04
9.1	16.97 16.91 16.96	0.847	8.7	75.7	6.25
10.1	17.00 17.03 17.06	0.852	7.7	59.3	5.58
11.1	17.22 17.37 17.38	0.866	6.7	44.9	5.03
12.1	17.78 17.72 17.75	0.888	5.7	32.5	4.49
13.1	18.57 18.59 18.47	0.927	4.7	22.1	4.04
14.1	19.78 19.90 19.75	0.991	3.7	13.7	3.69
15.1*	11.16 11.13 11.13	1.114	2.7	7.3	3.34
16.1*	13.25 13.40 13.50	1.338	1.7	2.9	3.04

* 15.1 va 16.1 sm da vaqt 10 sikl uchun.

Table 5: O'lchov natijalari jadvali

Metod (a) – chiziqli grafik usuli



To'g'ri chiziqli grafikdan hisoblash: qiyalik $\alpha = 0.04108 \pm 0.0007 \text{ s}^2/\text{sm}$, Y -kesishma $\beta = 3.10 \pm 0.05 \text{ s}^2 \cdot \text{sm}$.

$$g = \frac{4\pi^2}{\alpha} \quad \text{beradi} \quad g = (961 \pm 20) \text{ sm/s}^2 \quad (70)$$

$$\frac{\beta}{\alpha} = \frac{3.10}{0.04108} = 75.46 \text{ sm}^2 (\pm 2.5 \text{ sm}^2) \quad (71)$$

$$I_{CM} = (M + m) \frac{\beta}{\alpha} = (75.46)(M + m) \quad (72)$$

(4) tenglamadan:

$$I_{CM} = \frac{1}{3}M \left(\frac{L}{2} \right)^2 + M \left(x_{CM} - \frac{L}{2} \right)^2 + m(z - x_{CM})^2 \quad (73)$$

U holda

$$(75.46)(M + m) = 75.0M + 7.84M + m(z - 17.8)^2 \quad (74)$$

$$-7.38 \frac{M}{m} + 75.46 = (z - 17.8)^2 \quad (14)$$

Massalar markazi joylashuvi quyidagini beradi:

$$17.8(M + m) = 15.0M + mz \quad (75)$$

$$\frac{M}{m} = \frac{z - 17.8}{2.8} \quad (15)$$

(14) va (15) tenglamalardan:

$$-\frac{7.38}{2.8}(z - 17.8) + 75.46 = (z - 17.8)^2 \quad (76)$$

$$(z - 17.8) = 7.47 \quad (77)$$

va

$$z = 25.27 = 25.3 \pm 0.1 \text{ sm} \quad (78)$$

$$\frac{M}{m} = 2.68 = 2.7 \quad (79)$$

Xatolikni baholash

g uchun xatolikni toping:

(10) tenglamadan $g = \frac{4\pi^2}{\alpha}$,

$$\Delta g = \frac{\Delta \alpha}{\alpha} g = 16.3 \text{ sm/s}^2 \approx 20 \text{ sm/s}^2 \quad (80)$$

i) z uchun xatolikni toping:

Avval $r = \frac{\beta}{\alpha} = \frac{3.10}{0.04108} = 75.46 \text{ sm}^2$ uchun xatolikni toping.

$$\Delta r = \left(\frac{\Delta \alpha}{\alpha} + \frac{\Delta \beta}{\beta} \right) r = 2.5 \text{ sm}^2 \quad (81)$$

r dan keladigan xatolik eng katta hissa qo'shgani uchun ($\frac{\Delta r}{r} \sim 0.03$, $\frac{\Delta L}{L}$, $\frac{\Delta x_{cm}}{x_{cm}} \sim 0.005$), tahlilni soddalashtirish uchun faqat r dan xatolik tarqalishini baholaymiz. Endi $r_{\text{maks}} = r + \Delta r = 75.46 + 2.5 = 77.96$ ni qo'llaymiz. Tegishli kvadrat tenglama:

$$(z - 17.8)^2 + 1.743(z - 17.8) - 77.96 = 0 \quad (82)$$

Tegishli yechim $(z - 17.8)_{\text{maks}} = 7.55 \text{ sm}$.

Agar $r_{\text{min}} = r - \Delta r = 75.46 - 2.5 = 72.96$ ni ishlatsak, tegishli kvadrat tenglama:

$$(z - 17.8)^2 + 3.529(z - 17.8) - 72.96 = 0 \quad (83)$$

Tegishli yechim $(z - 17.8)_{\text{min}} = 6.96 \text{ sm}$.

Shunday qilib,

$$\Delta(z - 17.8) = \frac{7.55 - 6.96}{2} = 0.3 \text{ sm} \quad (84)$$

Eslatma: $\frac{\Delta(z-17.8)}{z-17.8} \sim 0.04$. Shuning uchun biz hali ham ΔL , Δx_{cm} dan xatolik tarqalishini e'tiborsiz qoldiramiz.

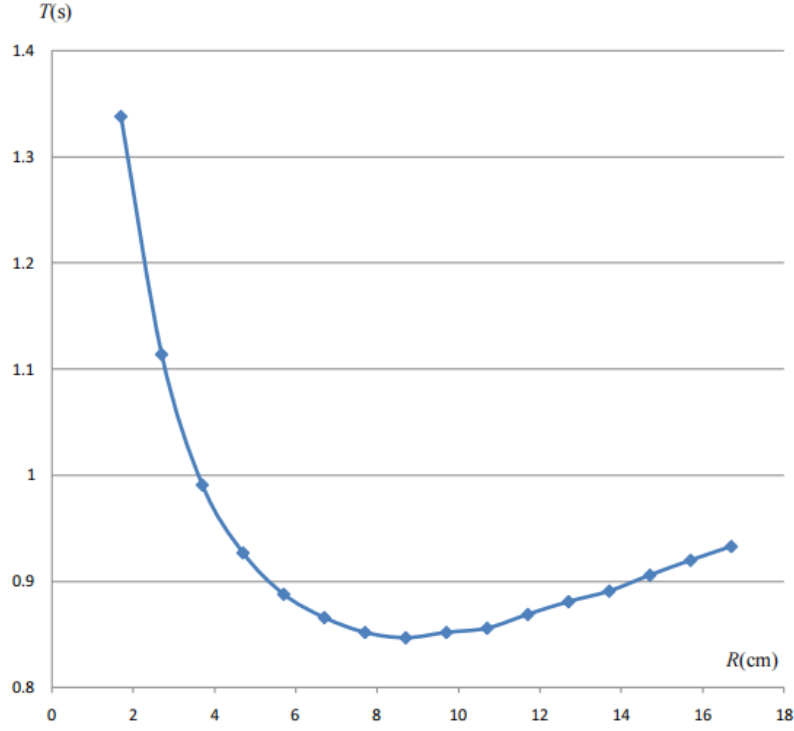
Δz ni $\Delta z \approx \Delta(z - 17.8) = 0.3 \text{ sm}$ deb baholash mumkin.

ii) $\frac{M}{m}$ uchun xatolikni toping:

Biz bilamizki $\frac{M}{m} = \frac{z-17.8}{2.8}$,

$$\Delta \left(\frac{M}{m} \right) = \frac{\Delta(z - 17.8)}{2.8} = 0.11 \quad (85)$$

Metod (b) – minimum nuqta egri chizig'i usuli



$T - R$ grafigidan hisoblash:

Minimum pozitsiyani ishlatib: $T = T_{\min}$ da $I_{CM} = (M + m)R_{\min}^2$ va $g = \frac{8\pi^2 R_{\min}}{T_{\min}^2}$.

Grafikdan: $R_{\min} = 8.9 \pm 0.2$ sm va $T_{\min} = 0.846 \pm 0.005$ s.

$$\therefore g = 982 \pm 40 \text{ sm/s}^2 \quad (86)$$

$$I_{CM} = (M + m)(8.9)^2 = (79.21)(M + m) \quad (16)$$

(14), (15), (16) tenglamalardan:

$$(79.21)(M + m) = 75.0M + 7.84M + m(z - 17.8)^2 \quad (87)$$

$$-3.63M + 79.21m = m(z - 17.8)^2 \quad (88)$$

$$(x - 17.8)^2 + \frac{3.63}{2.8}(x - 17.8) - 79.21 = 0 \quad (89)$$

$$(z - 17.8) = 8.28 \quad (90)$$

va

$$z = 26.08 = 26.1 \pm 0.7 \text{ sm} \quad (91)$$

$$\frac{M}{m} = 2.95 = 3.0 \pm 0.3 \quad (92)$$

Xatolikni baholash (metod b)

i) g uchun xatolikni toping:

Minimum pozitsiyani ishlatib: $g = \frac{8\pi^2 R_{\min}}{T_{\min}^2}$, bizda

$$\Delta g = \left(\frac{\Delta R_{\min}}{R_{\min}} + 2 \frac{\Delta T_{\min}}{T_{\min}} \right) g = 34 \approx 30 \text{ sm/s}^2 \quad (93)$$

ii) z uchun xatolikni toping:

Avval $r = R_{\min}^2 = 79.21 \text{ sm}^2$ uchun xatolikni toping.

$$\Delta r = 2R_{\min} \Delta R_{\min} = 3.56 \text{ sm}^2 \quad (94)$$

Bu r 1-qismdagi r ga ekvivalentdir. Shunday qilib, xuddi shu xatolik tahliliga amal qilish mumkin.

Natijada,

$$z = 26.08 \approx 26.1 \text{ sm} \quad (95)$$

$$\Delta z = 0.8 \text{ sm} \quad (96)$$

iii) $\frac{M}{m}$ uchun xatolikni toping:

I qismdagi kabi tahlilga amal qilib, topdik:

$$\frac{M}{m} = 2.96; \quad \Delta \left(\frac{M}{m} \right) = 0.15 \quad (97)$$

ESLATMA: Ushbu minimum egri chizig'i usuli odatdagi to'g'ri chiziqli grafik usuli kabi aniq emas.